A Short Look at a Tangent Laced Figure; Evolving From the Byzantine Tradition. Khirbat al Mafjar, Jericho.

Photo: Flickr; Alex Brey CC BY-SA 2.0. https://creativecommons.org/licenses/by-sa/2.0/

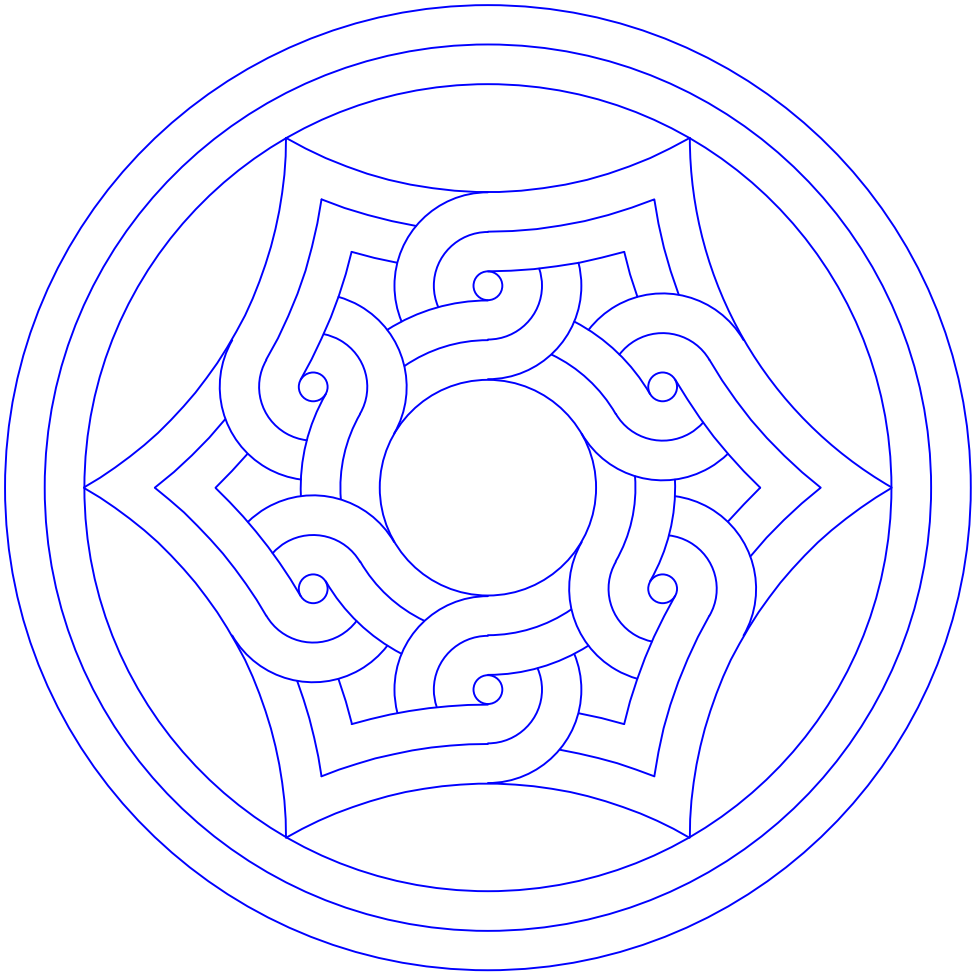
This very fine interlaced pattern is from a palace at Jericho, Khirbat al Mafjar, also known as Hisham's palace, Qaṣr Hishām.

The date for the palace is certain within a few years; it is Umayyad from the first half of the 8th Century CE.
This date is at the very beginning of what will become the unique Islamic geometric tradition.

It is interesting as a pattern that sits comfortable between the older Byzantine tradition of the area and what will be come the uniquely Islamic tradition.

.

It is very closely related to Roman / Byzantine mosaic patterns, also found at Khirbat al Mafjar. At the same time it is uniquely something new.



It is also interesting as a tangent interlaced design. These are found throughout Byzantine art and into Islamic art of all following periods.

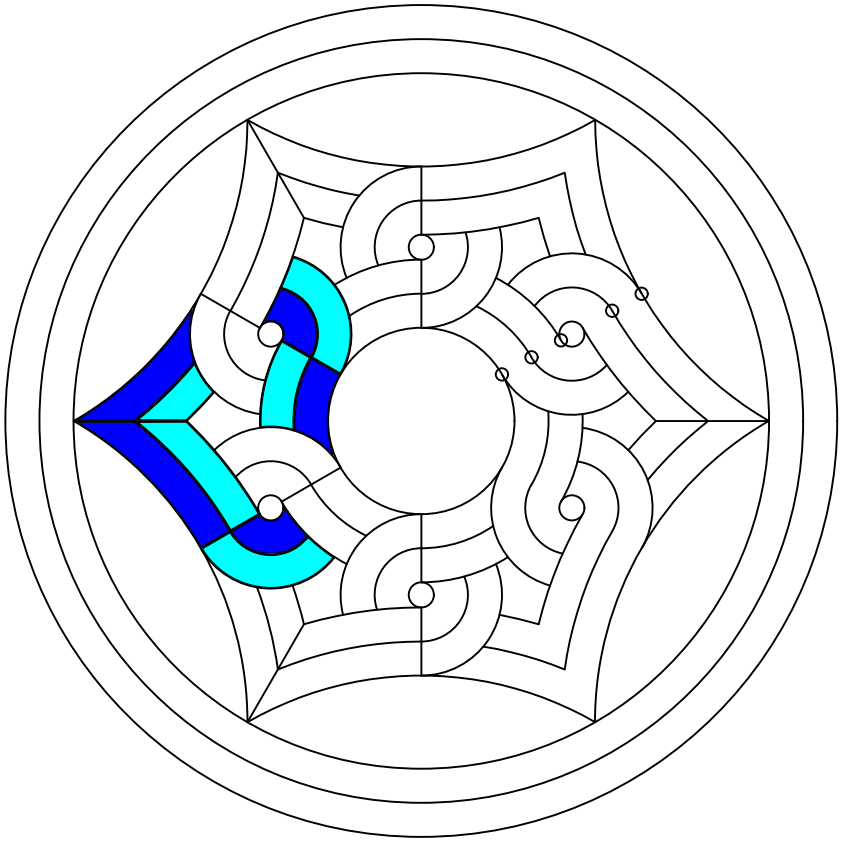
.

The pattern is formed only by smooth arcs of circles which meet at their tangent points

.

These are a challenge to draw well, but can be quite impressive.

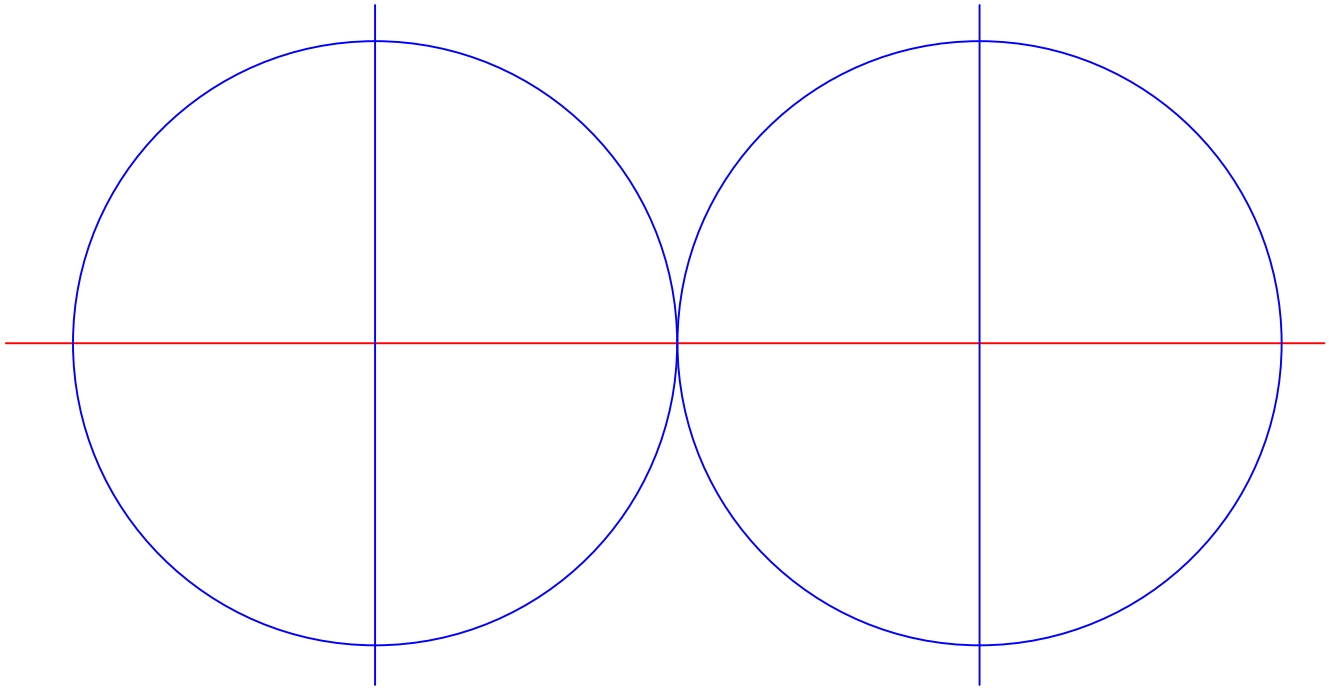
First, Some definitions.



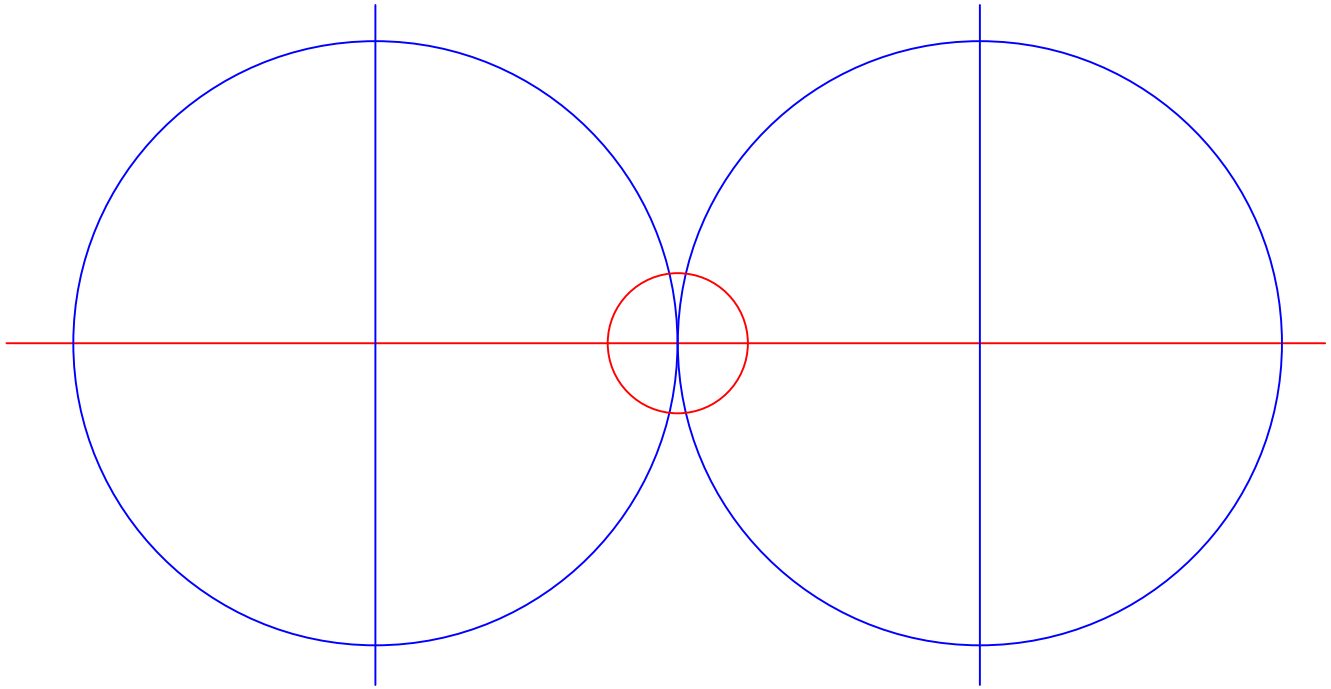
Every tangent point, marked on the top right side, is where two arcs of circles meet and change direction in the pattern.
A smooth tangent laced figure constantly reverses and changes directions

.

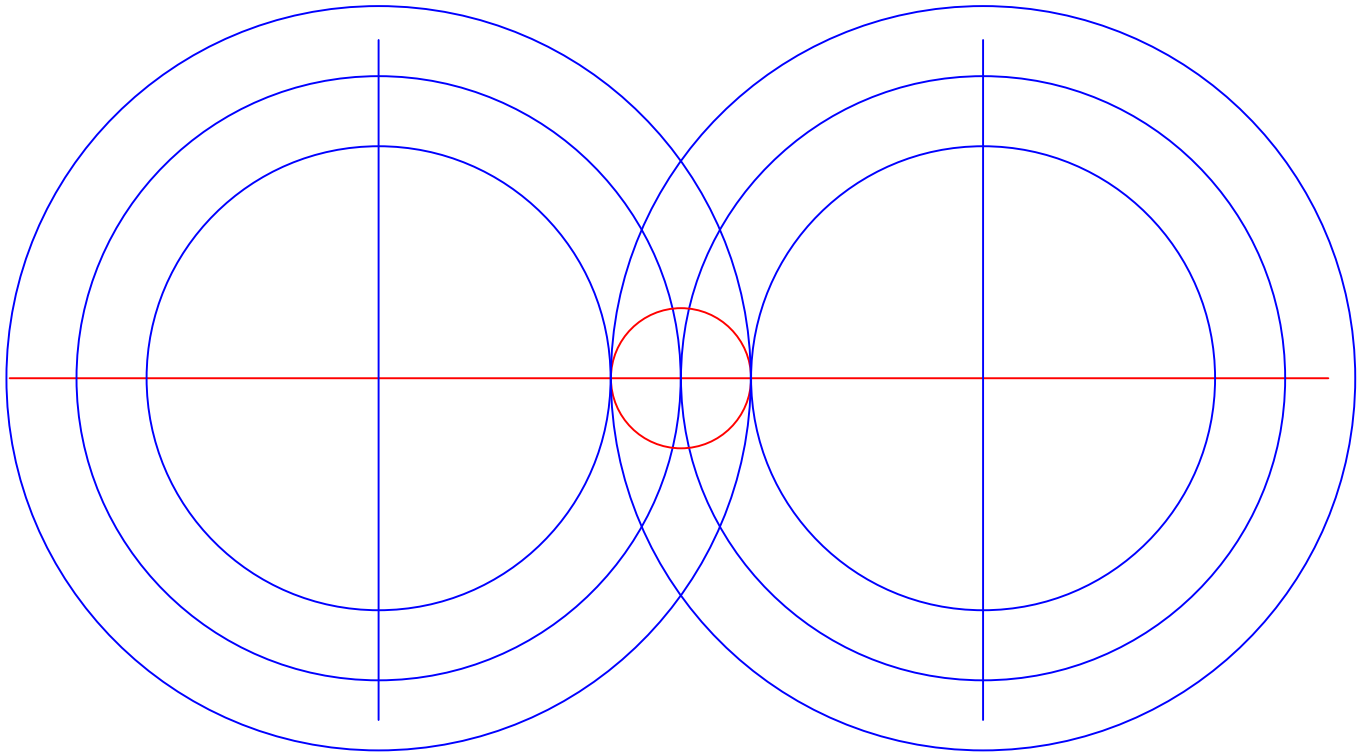
The challenge is to locate the tangent point exactly to get smooth curves.



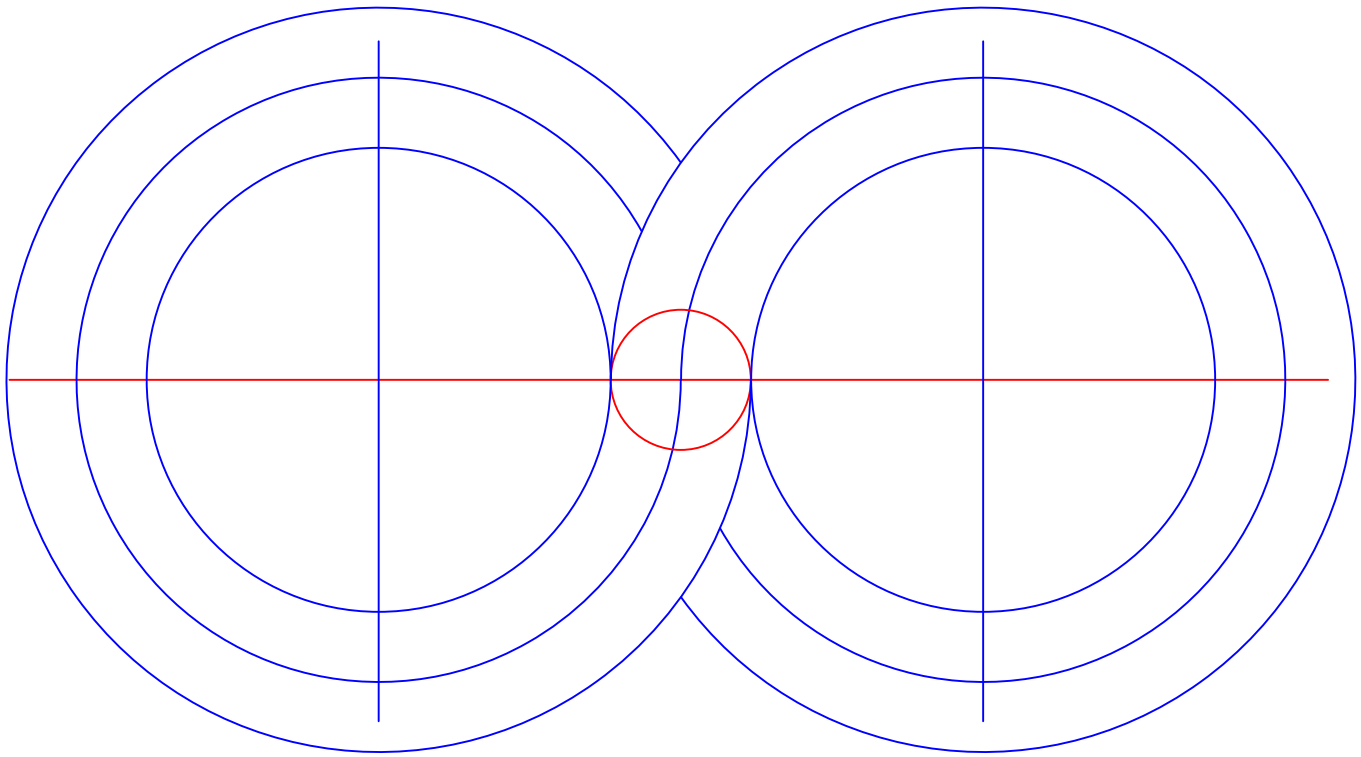
"Tangent circles" (or other curves) touch and overlap at a single point.
This means that they have a single shared radius, in red.
.
This radius is how we locate tangent points.



The Khirbat al Mafjar figure has a thick interlacelace defining the pattern.
The easy way to define this is by a circle at a tangent point.
.
Where this red circle crosses the red radius can define the width of the interlace.



The inside and outside of the interlace are defined by two concentric circles from each center.
Note that the inside of one circle becomes the outside of the tangent circle after it crosses the shared radius.
.
For just this very simple figure we have 6 circles changing directions at 3 points.
These get complicated.



The interlacing circles obey the same over under alternate rules as any other interlace figure.

.

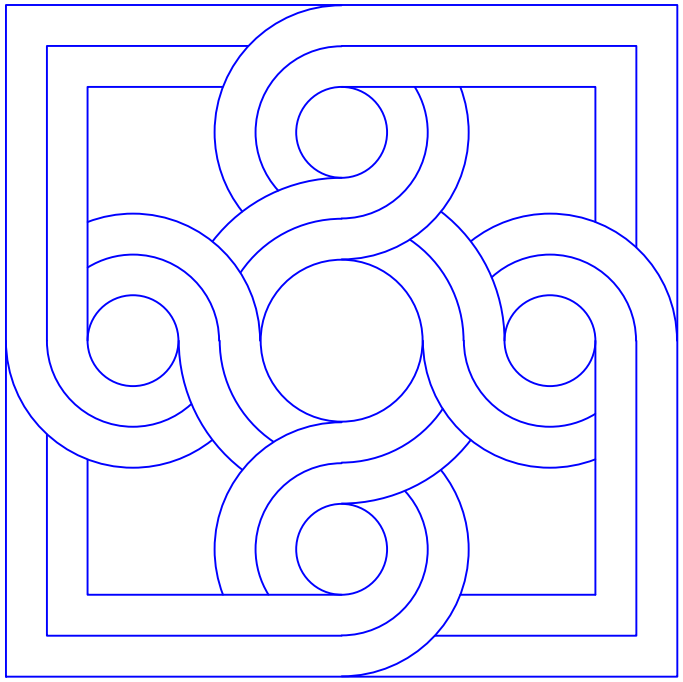
Cleaning up the crossing, we are done.

.

Easy, Right? But not very exciting.

A Byzantine Classic

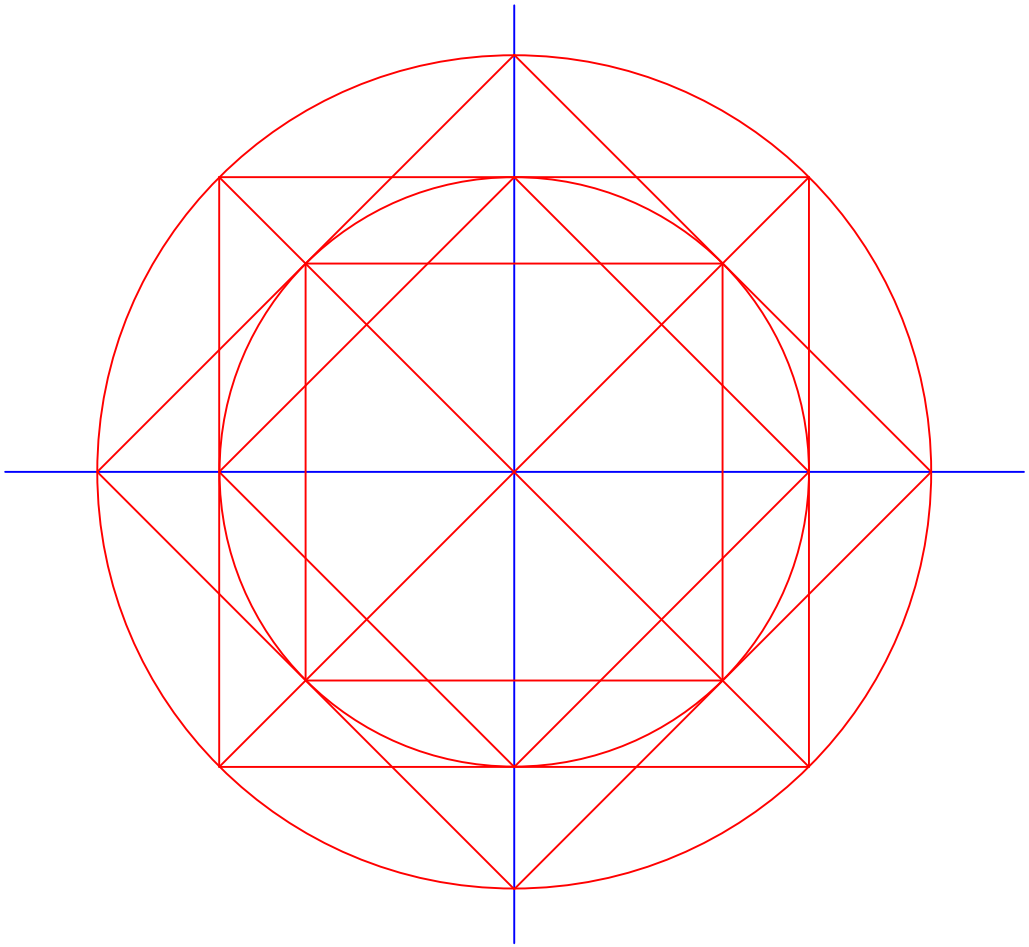
More complex, and more interesting, patterns are not too much harder to draw.
This pattern has 15 circles and 24 tangent points, but it is not difficult.



This pattern is found in Byzantine mosaics spanning several centuries,
and it is even found at Khirbat al Mefjar in the mosaic floors as an infinite repeat.
It reappears in Europe in the Renaissance, a period spanning over 1000 years.
A classic

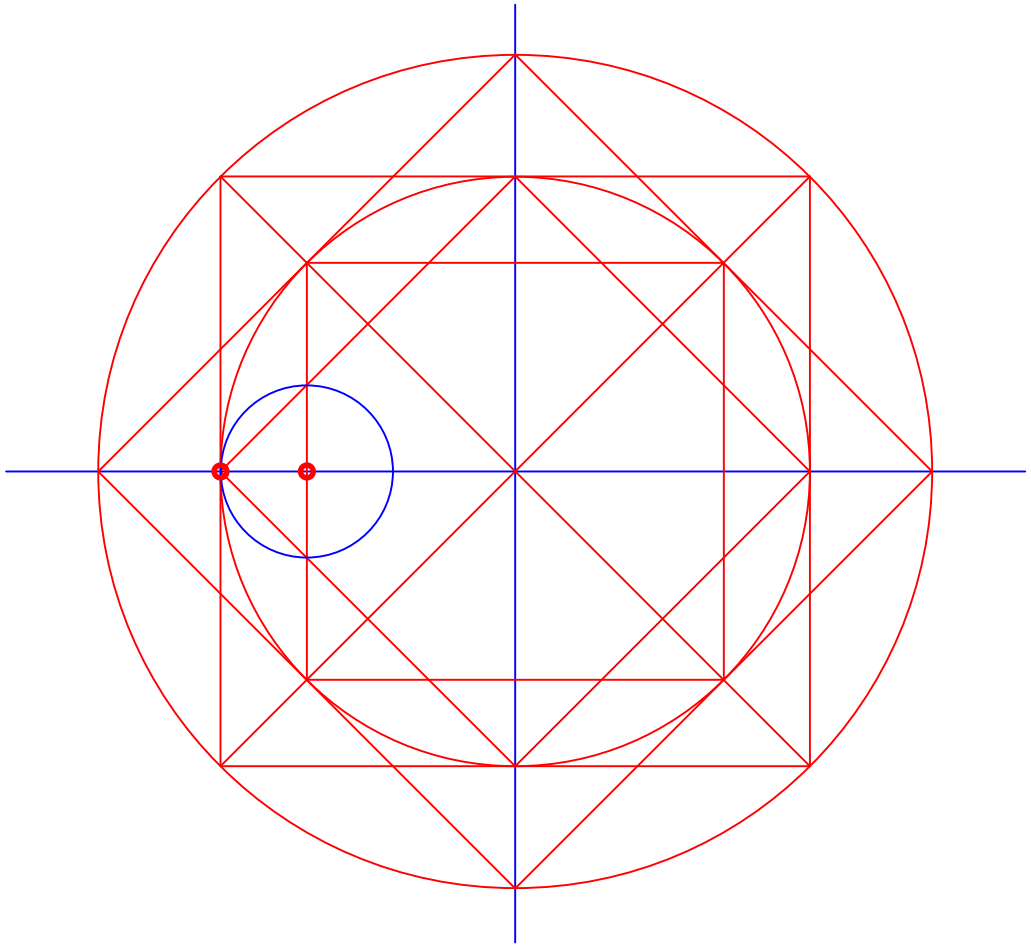
Some Byzantine Interlace Mosaic at Saints Cosmas and Damian, in Jerash, 533 CE
The same pattern appears at Khirbat al Mafjar converted to an infinite repeat,

Thsi pattern has 4 circles around a center circle. The fairly obvious place to start is with a square.



I started with this basic layout: Two squares drawn inside a circle. These define a new circle.
Two squares are drawn inside the inner circle. This is a common base for a pattern.

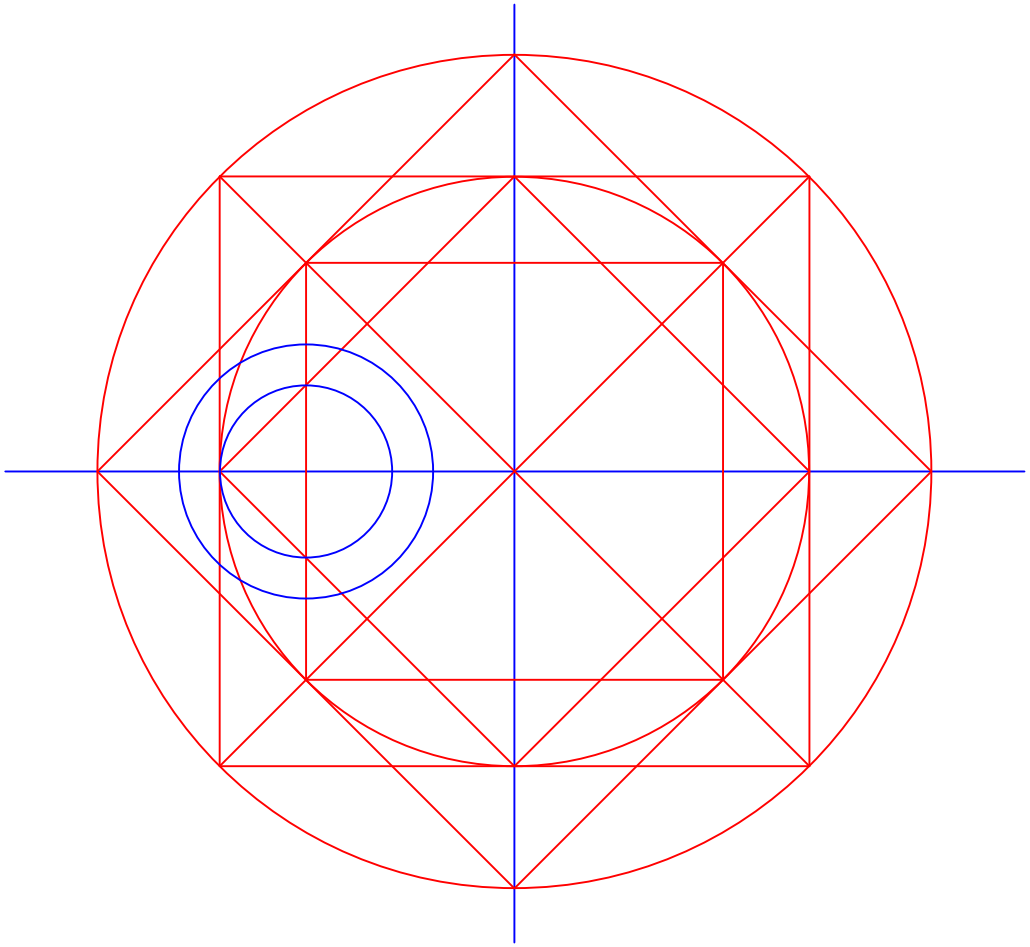
This is not a completely "Determinate Pattern." There are several choices to make within some fairly wide ranges. The final pattern will vary quite a bit, based on your choices.



The first choice is the size of the outer circles. This circle will determine the size of the inner circle and the overall proportions of the pattern.

This will be the centerline of the pattern. It will give proportions similar to many historic examples.

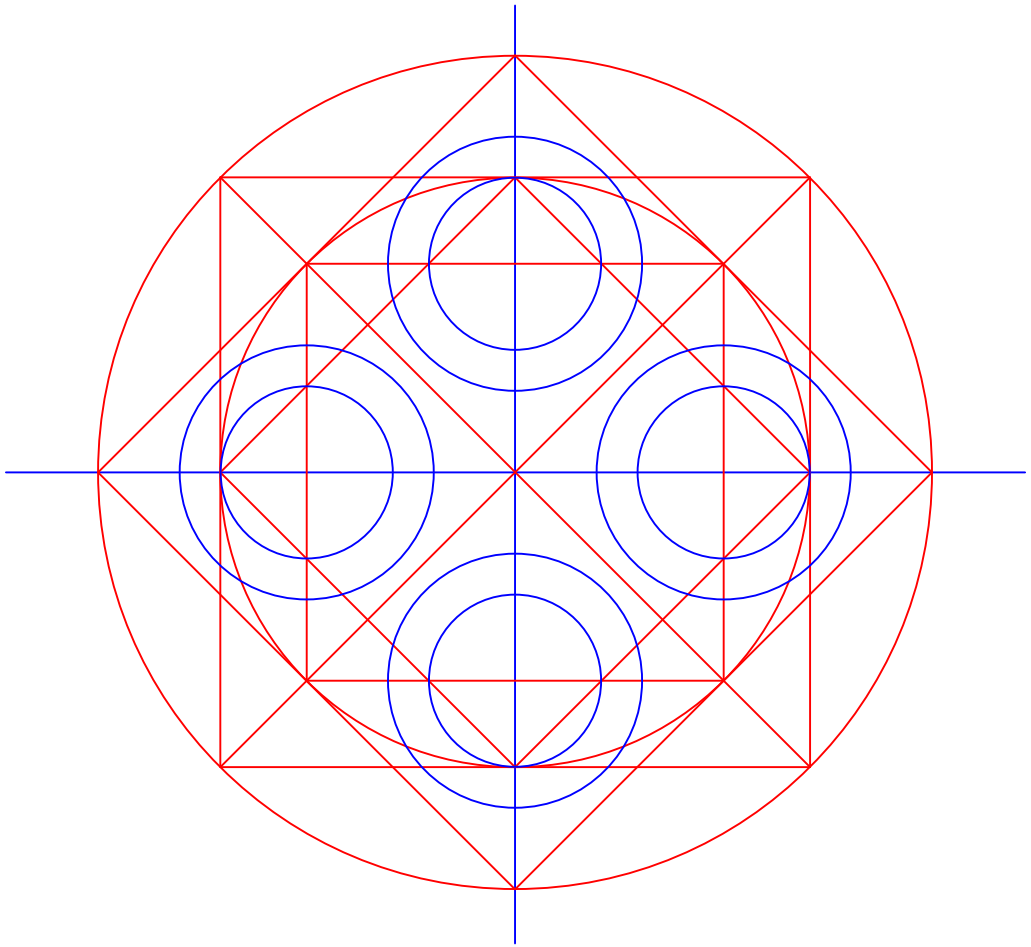
The next free choice is how wide the lacing of the pattern will be.



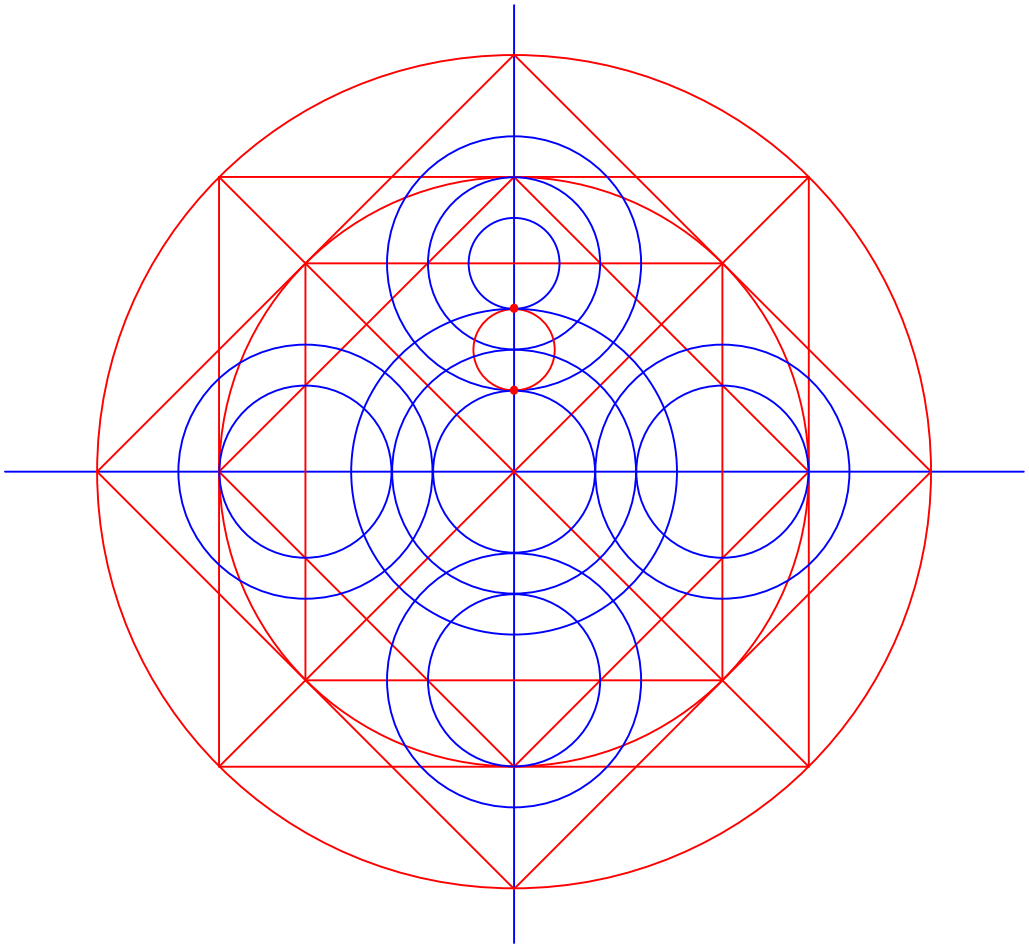
One side, inside or outside, is chosen. The second side is decided by symmetry in the next steps.

This radius was chosen to give pleasing proportions similar to historic patterns. I just picked a radius.
I don't normally do this. Normally I would use a point or an intersection in the definition of the layout.
I will show where this could be done in later steps.

Filling in the four corners gives this. At this point, the entire pattern is defined.



The chosen diameter for the outside of the small circle defines the inside of the center.

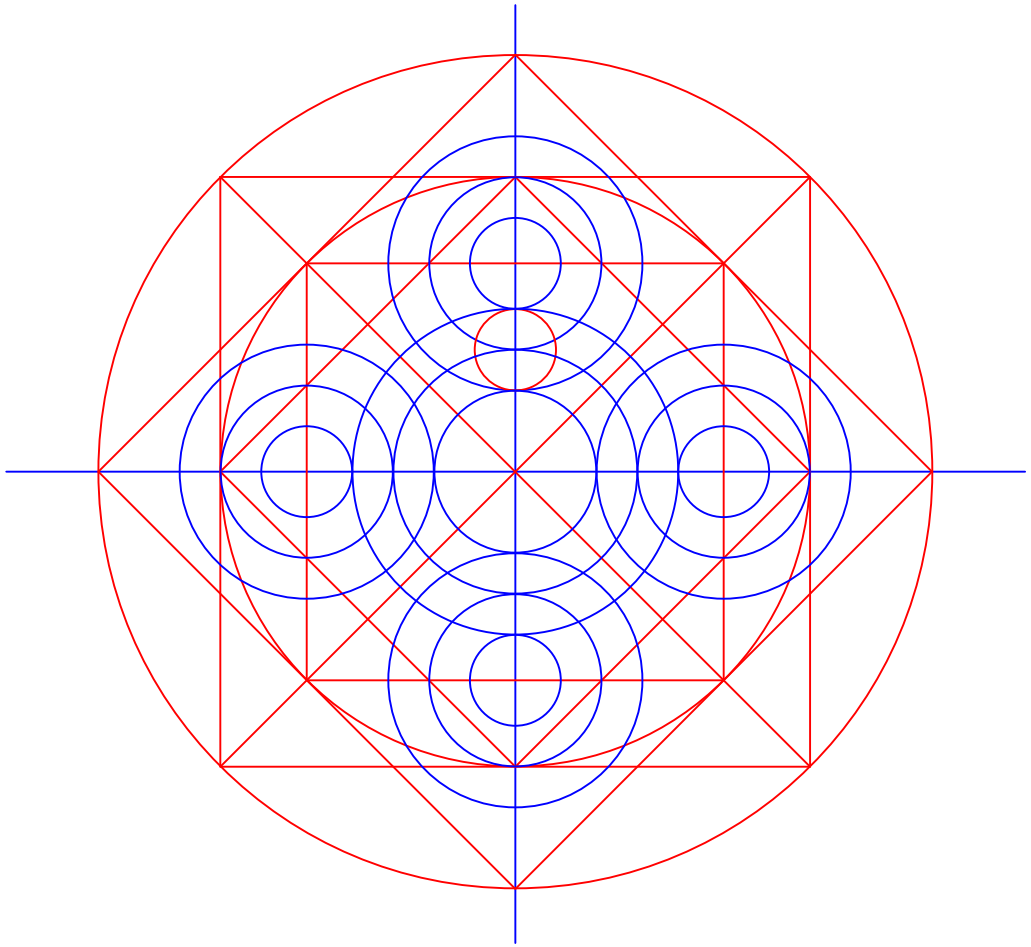


Remember that the outside of one circle becomes the inside of the next tangent circle when it is laced.

The lacing width is set, but the width needs to be transferred to the other side of the pattern center-line.
The small red circle is drawn around the center-line on a shared radius- with the radius defined by the existing lacing border where it crosses the shared radius.
The new point at the top of the circle defines the symmetric lacing- where it crosses the shared radius.

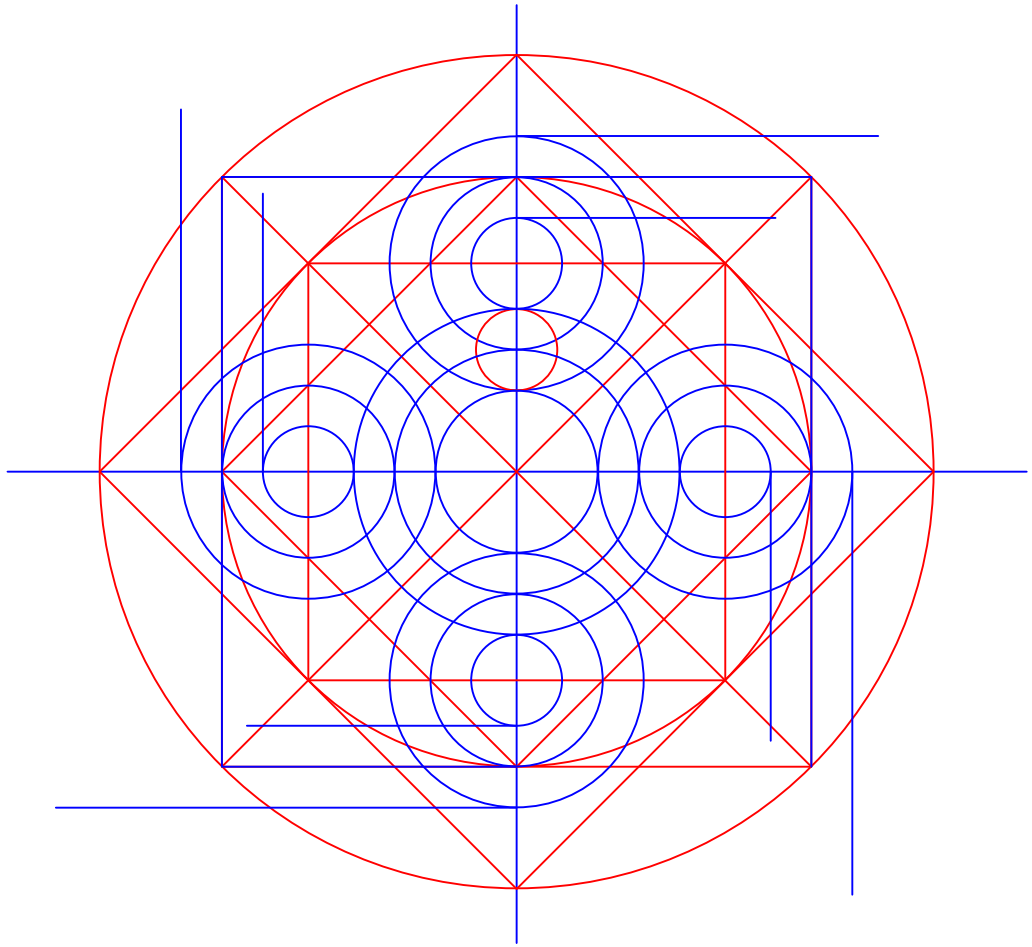
Tangent dimensions and intersections always defined on the shared radius.

Transferring the new lacing radius around, we are here.



Classic Byzantine examples of this pattern are usually finished as an isolated pattern, very much like the Khirabt al Mafjar oculus pattern.

To finish this pattern as an isolated figure, the lacing is continued around the original layout square.



A detail of the layout is on the next page.

The points to define the lacing at the corners of the pattern are just the same two circles, inside and outside diameter of the lacing, transferred out to the corners of the layout square.

.

You only need to mark the intersection on the extended red line, but I included the larger arc to show one place where you could define the lacing width based on the layout structure.

.

I could have use the indicated point as a definition of the width.

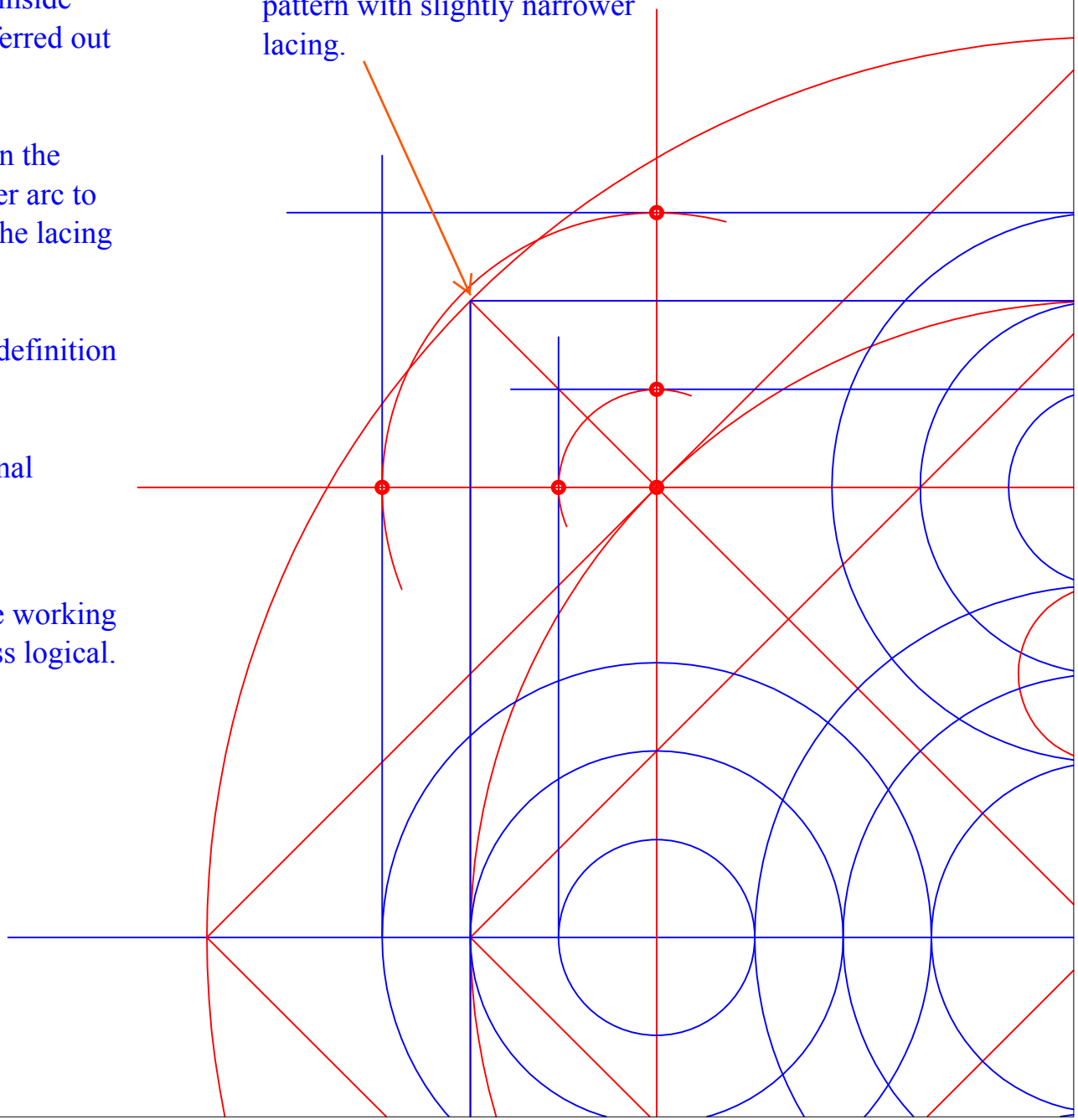
.

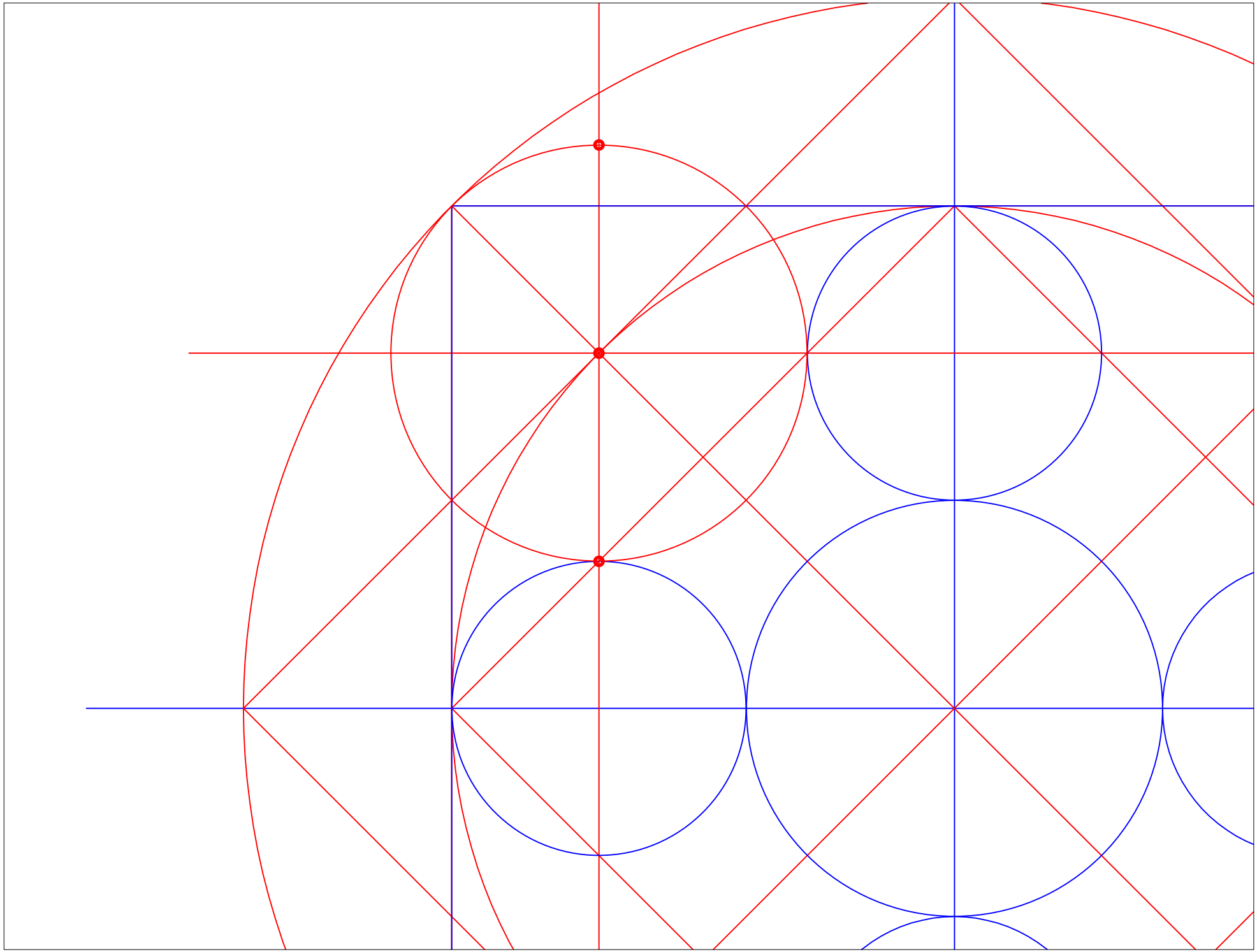
I chose a lace just a little wider for personal preference.

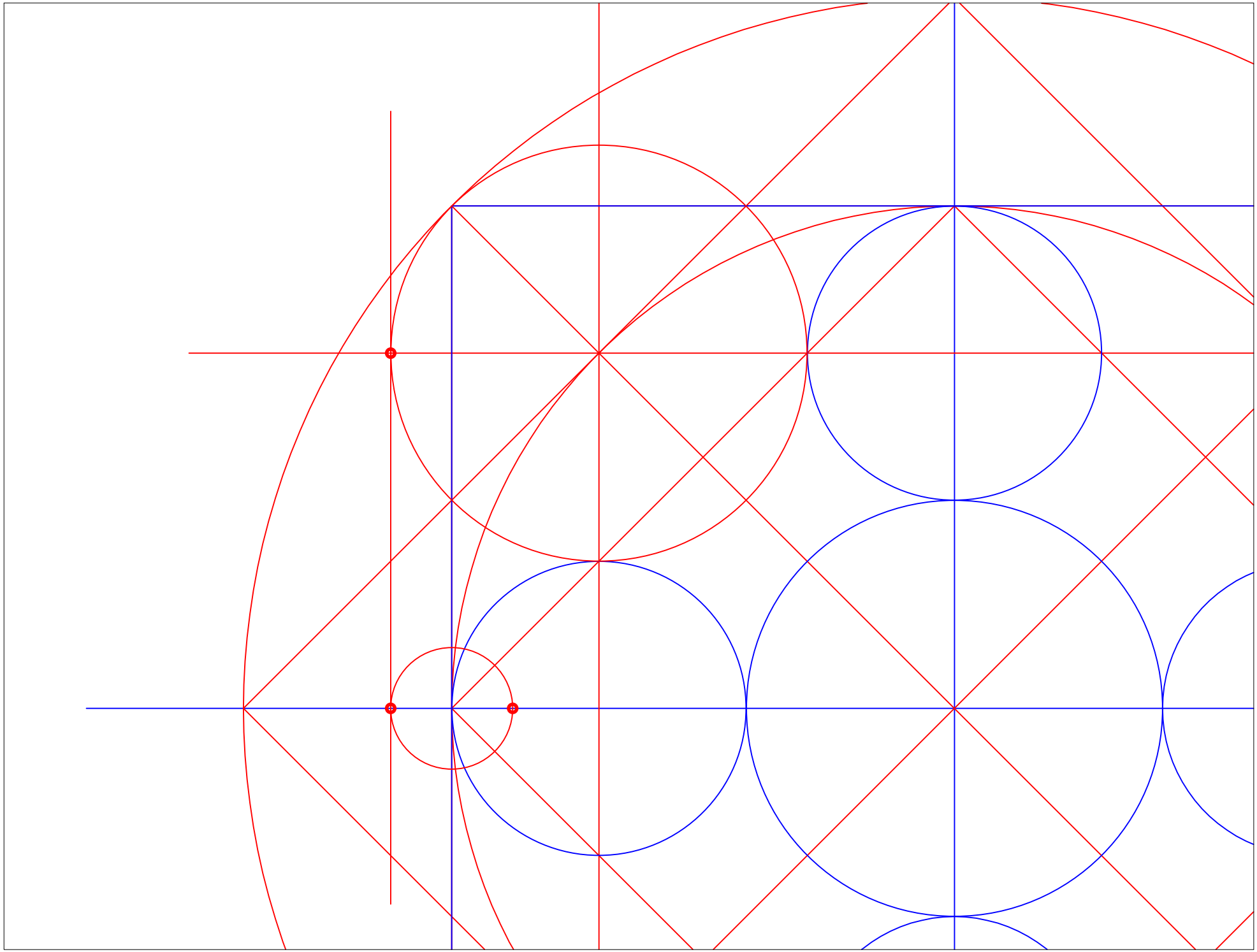
I decided not to use this since it would be working backwards into the pattern and appear less logical.

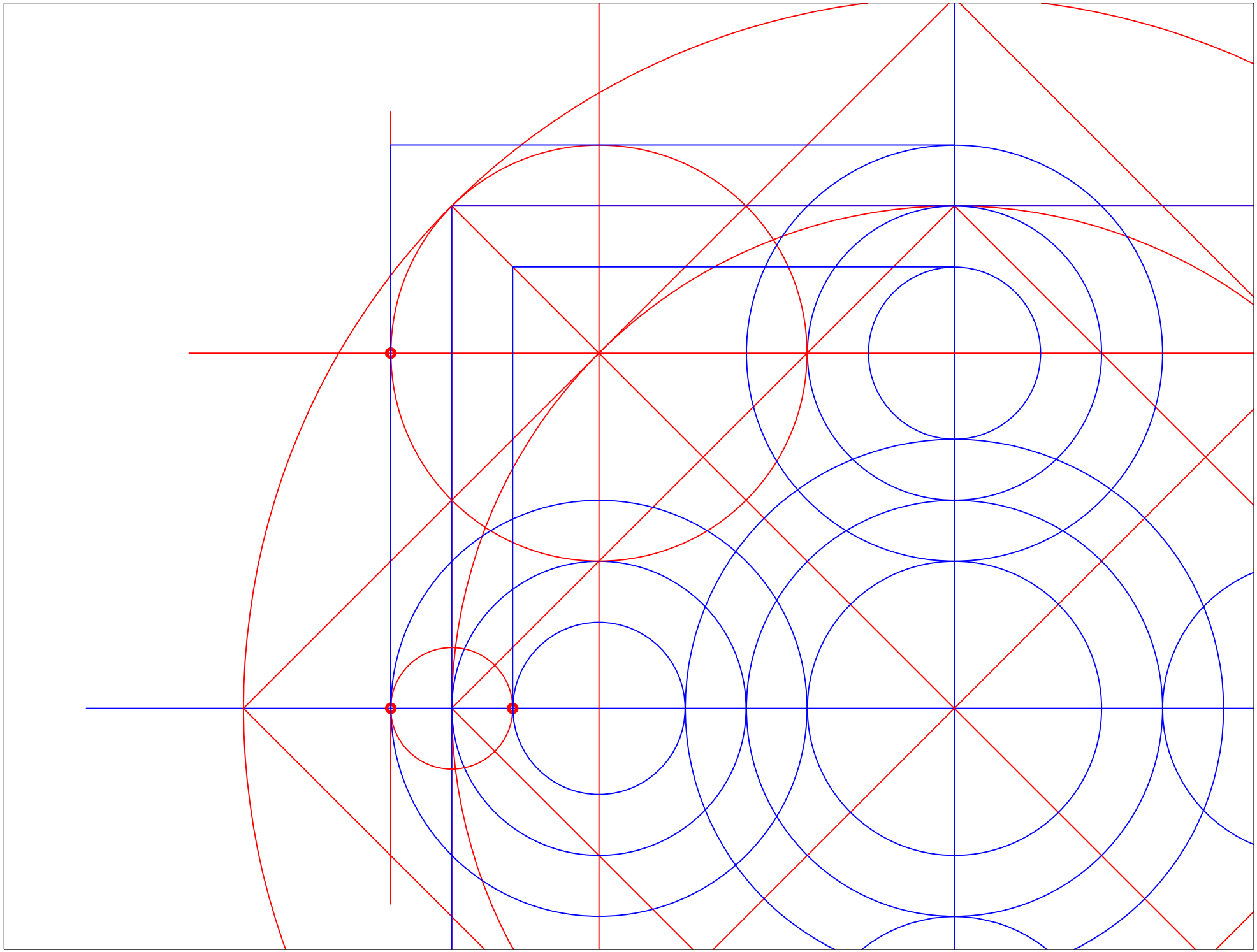
A short summary of how this would work is shown on the next three pages.

I could have used this to define lacing and had a very similar pattern with slightly narrower lacing.

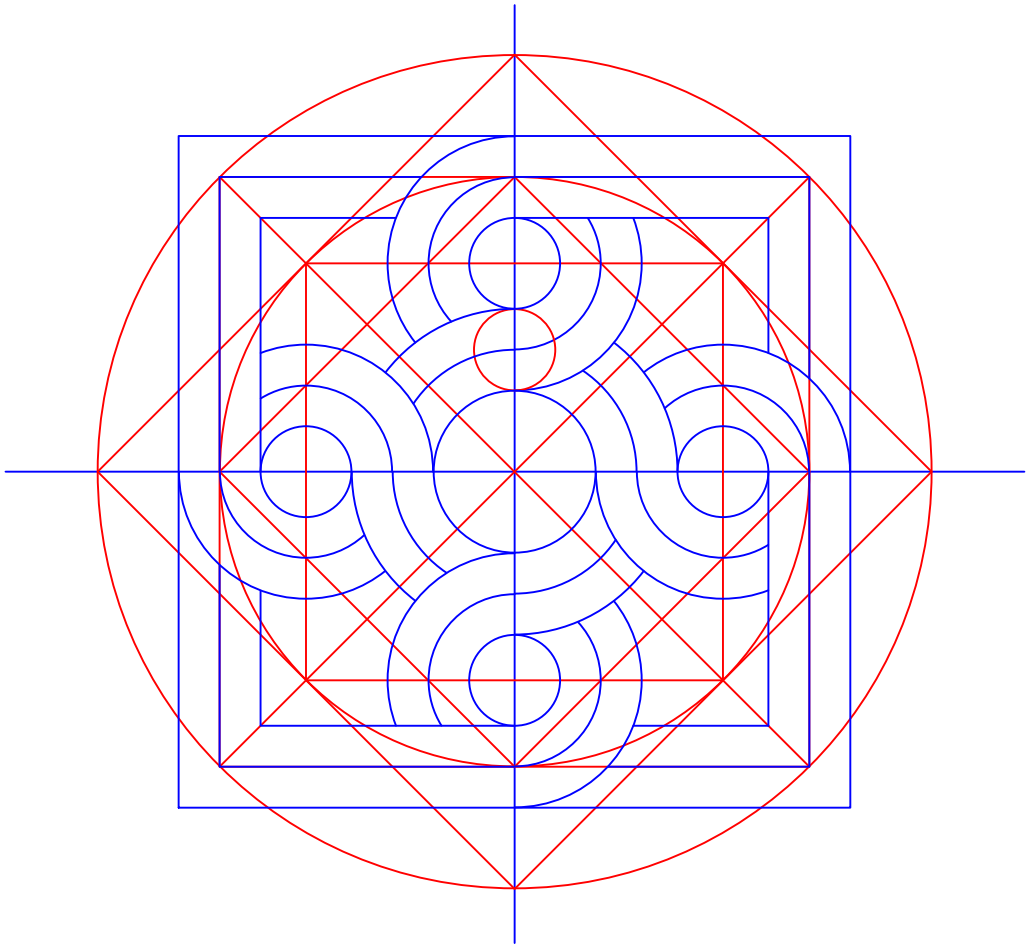


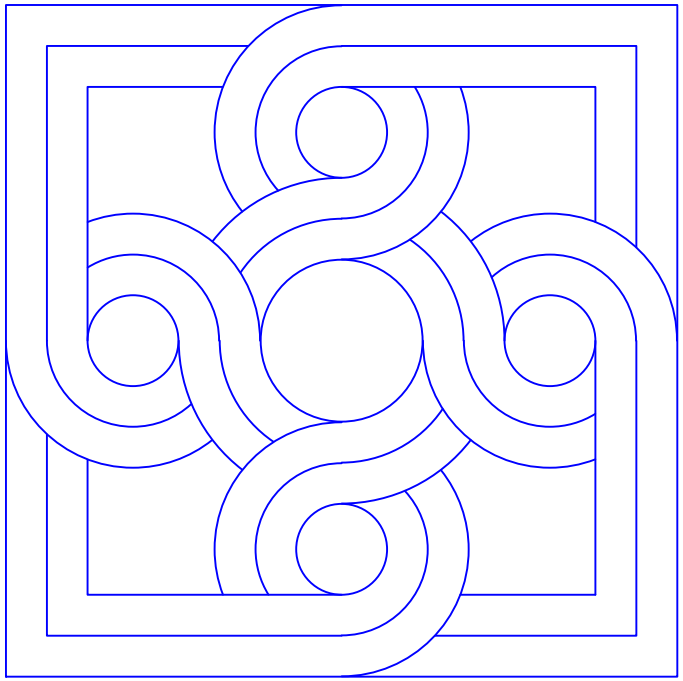


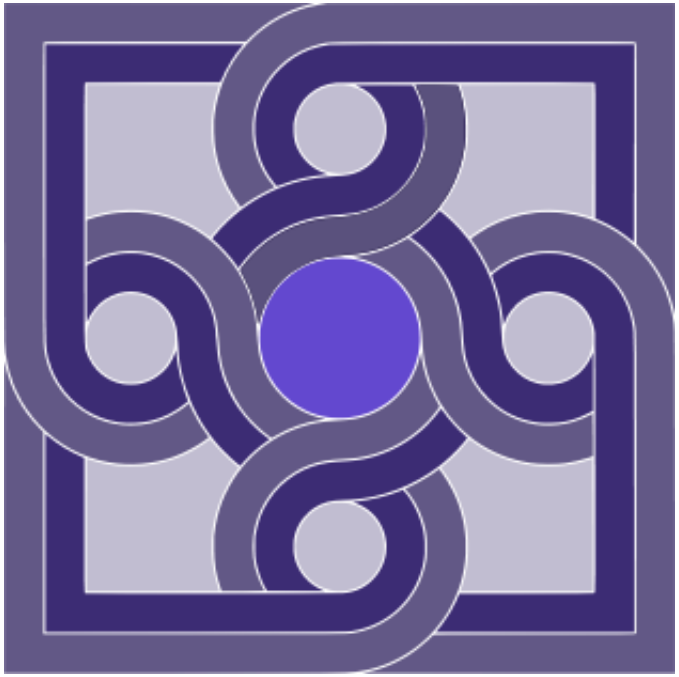


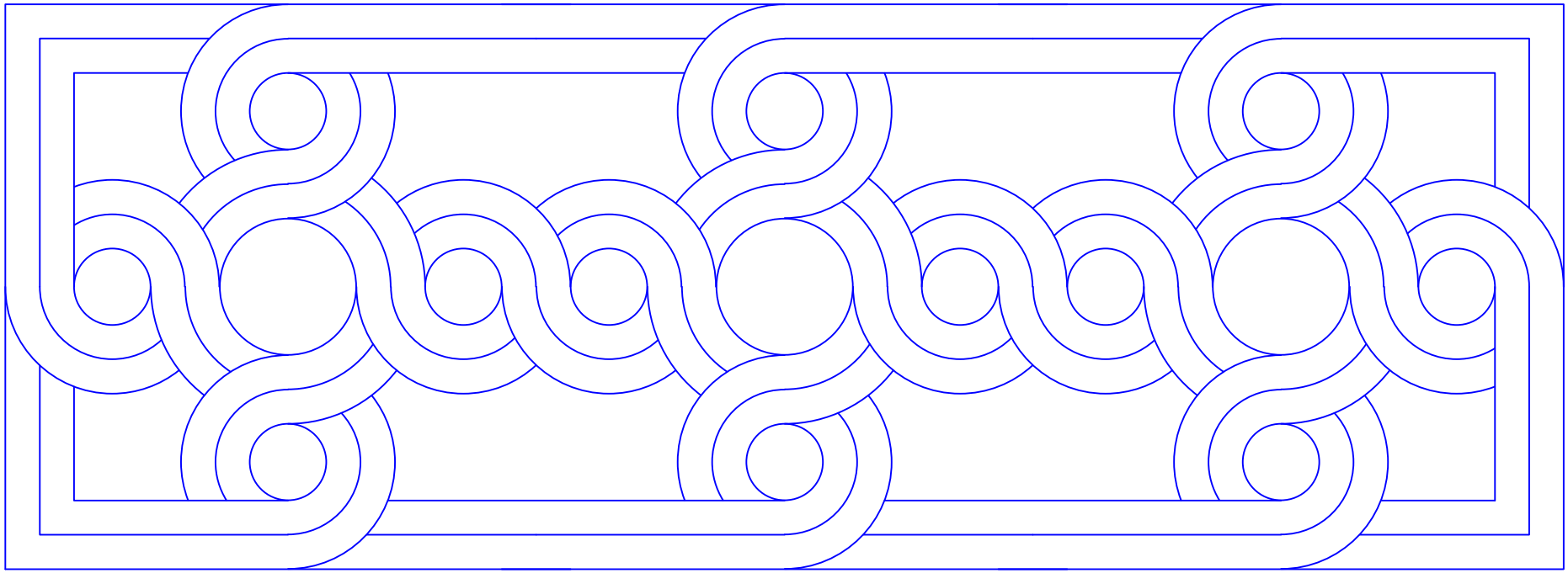


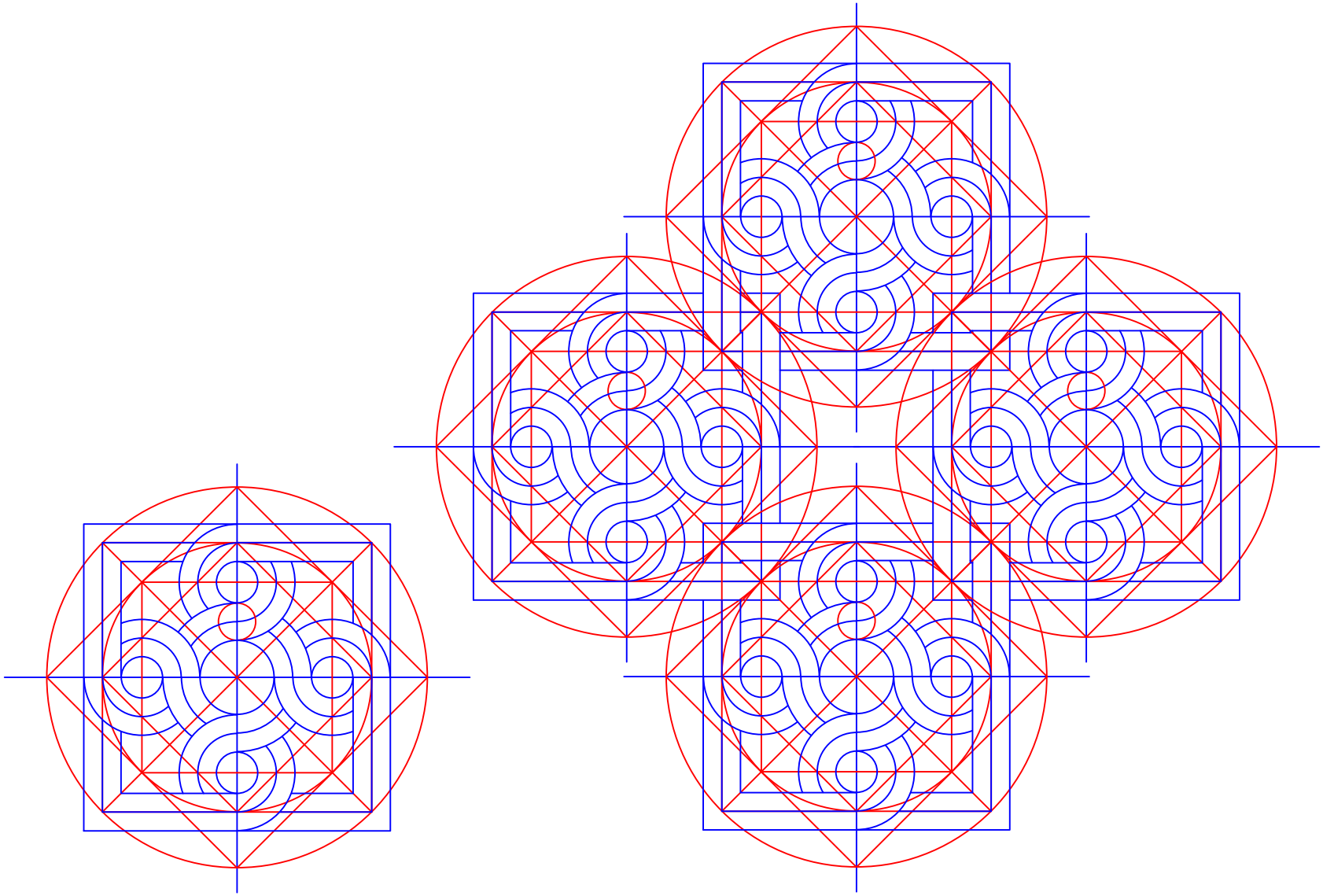
The final isolated figure, with its full layout, looks like this.

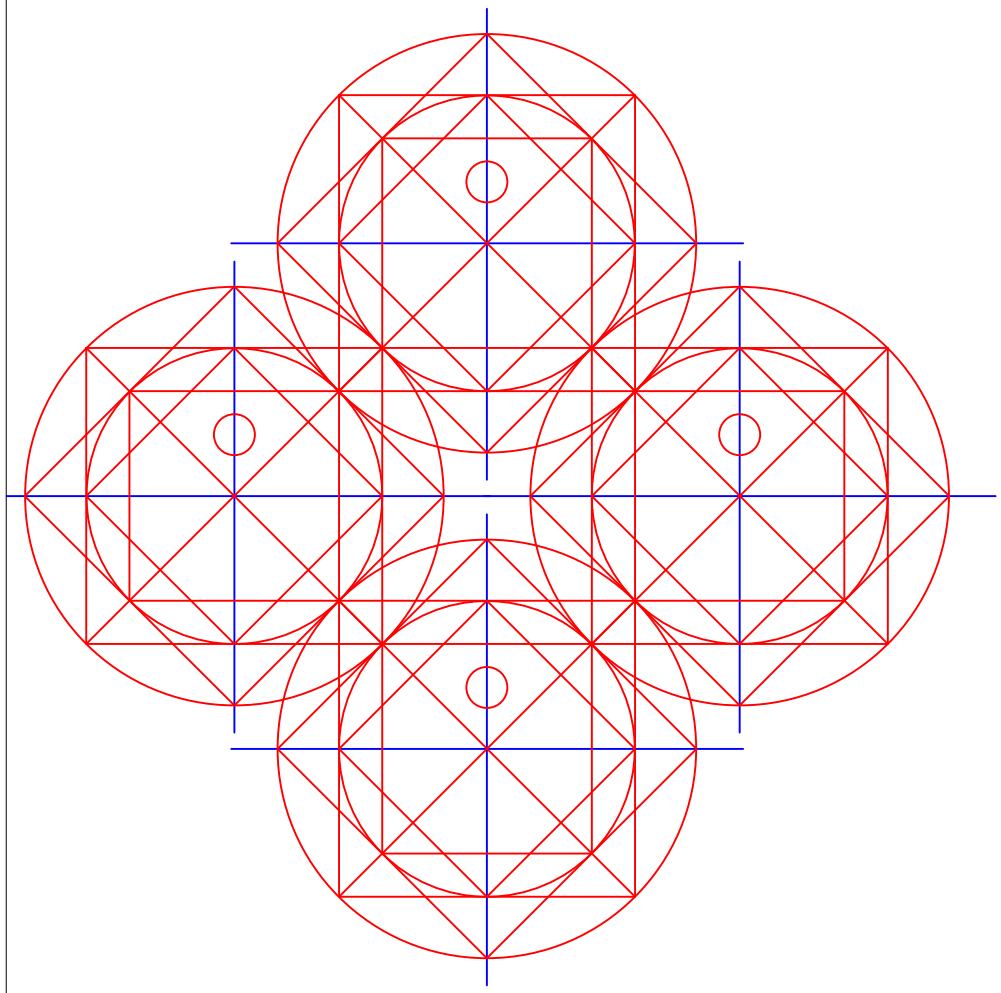


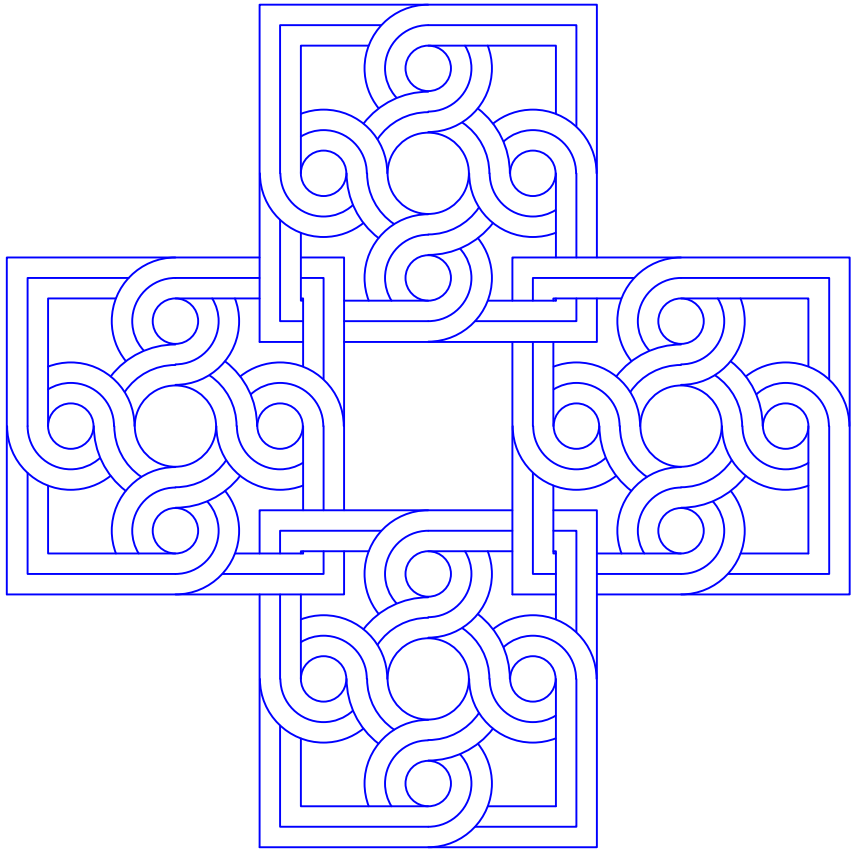


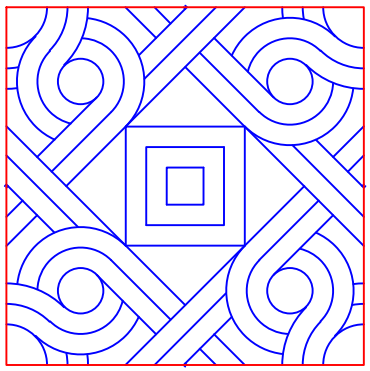


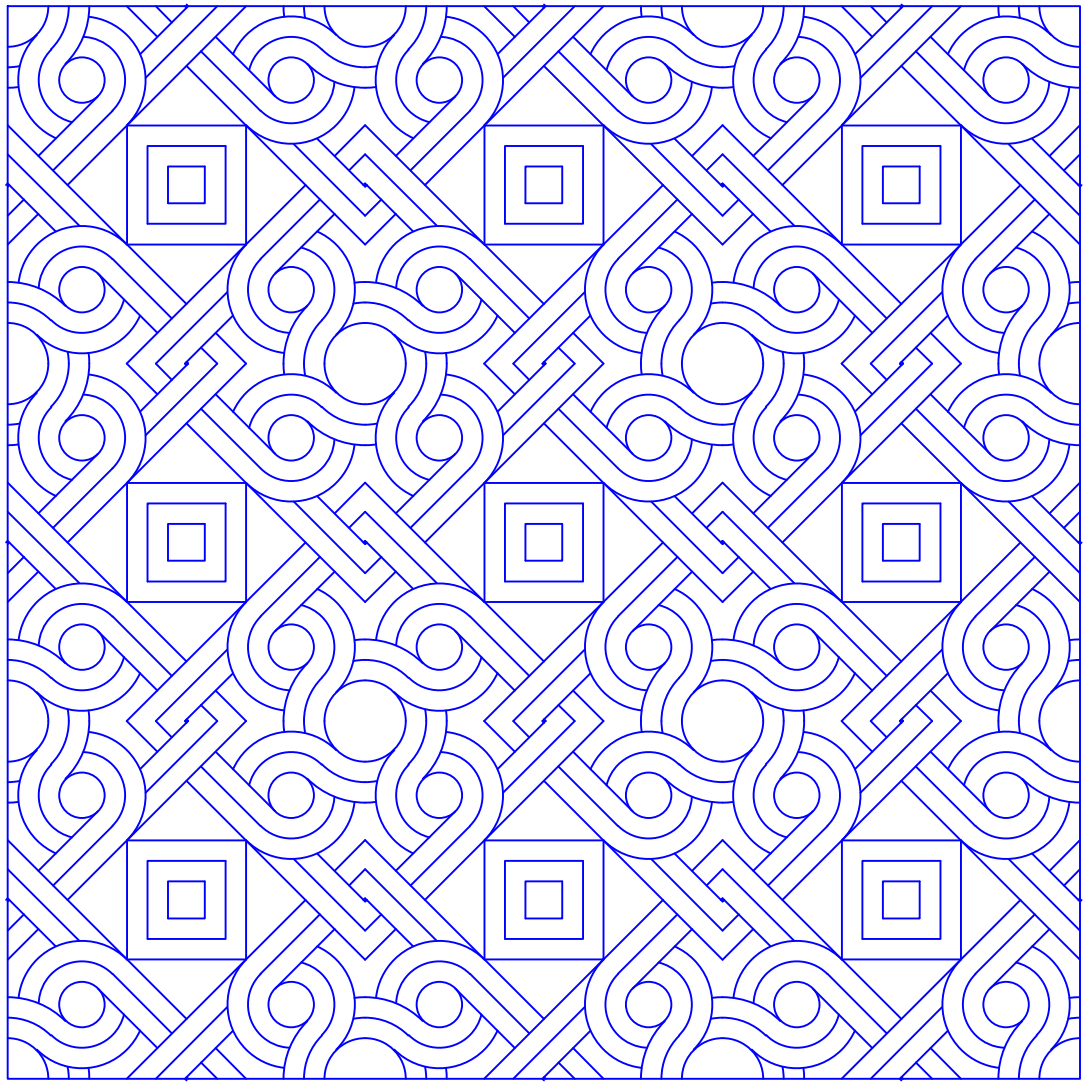


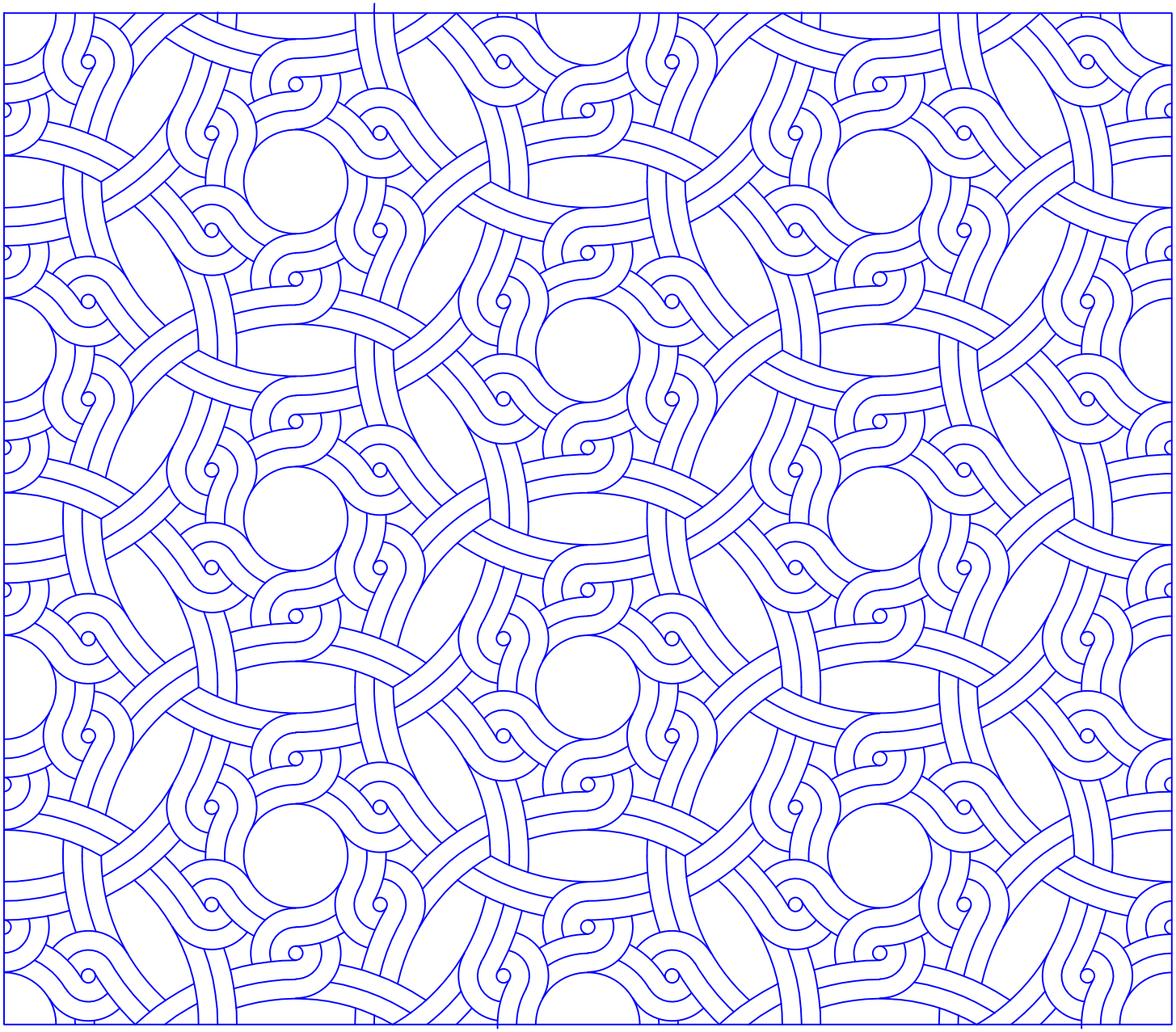










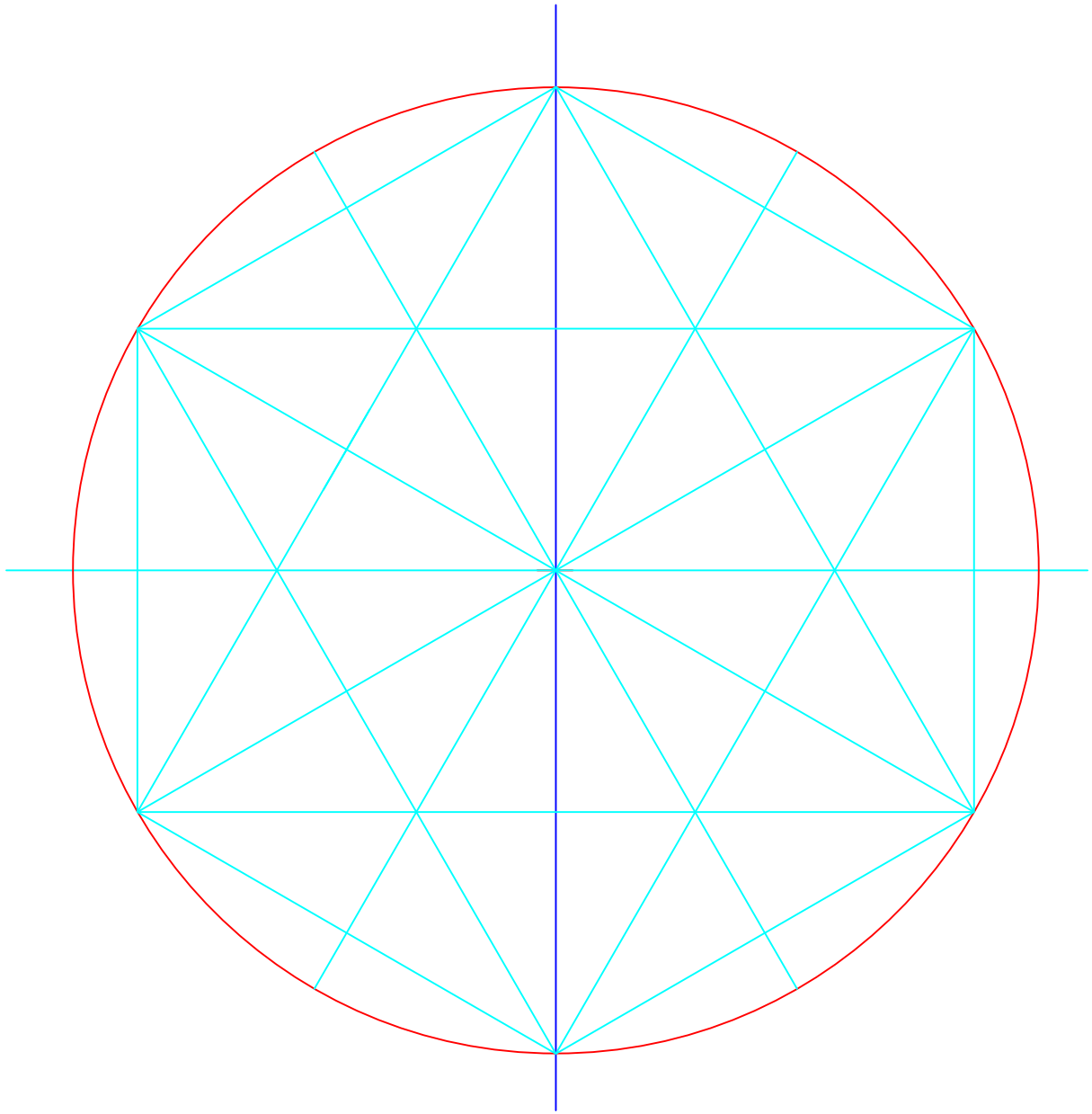
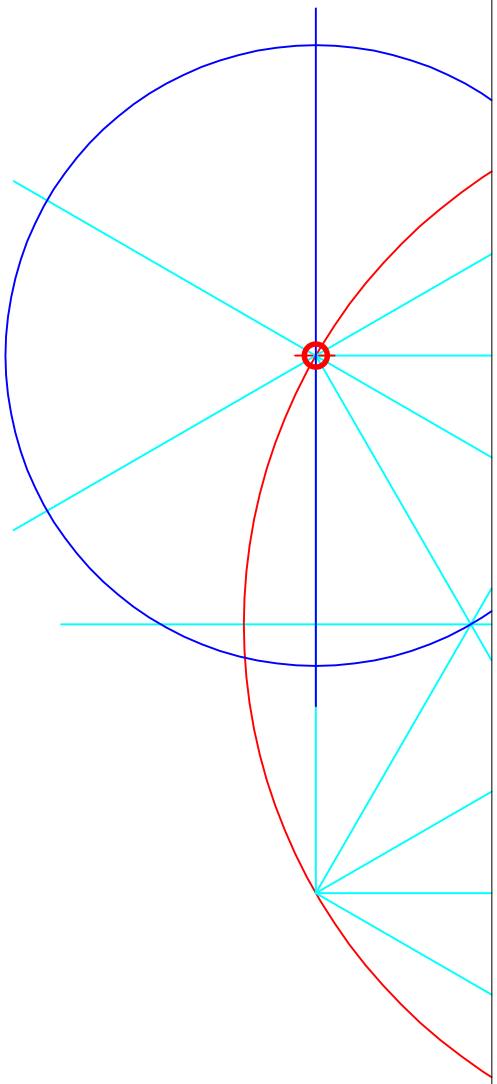
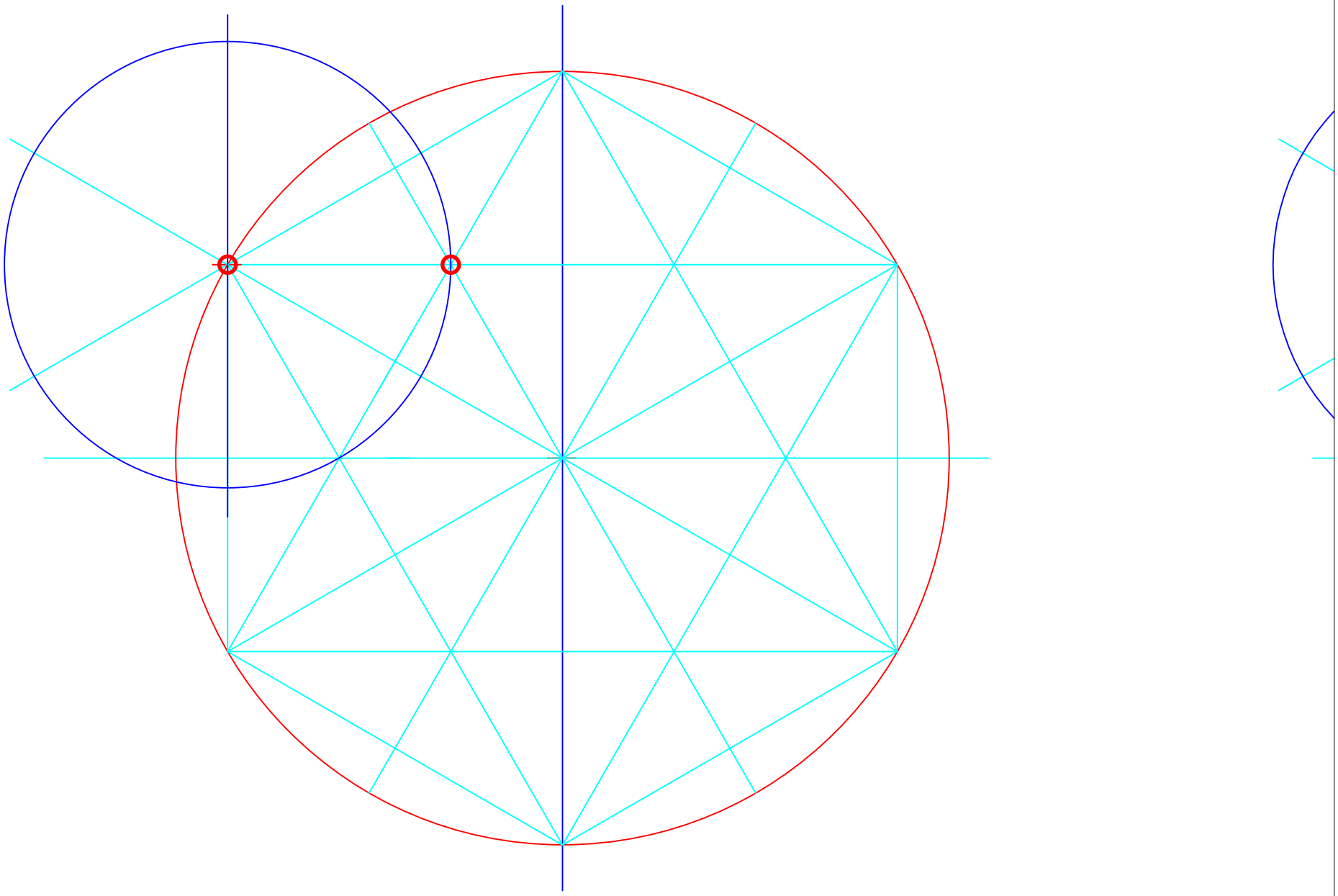
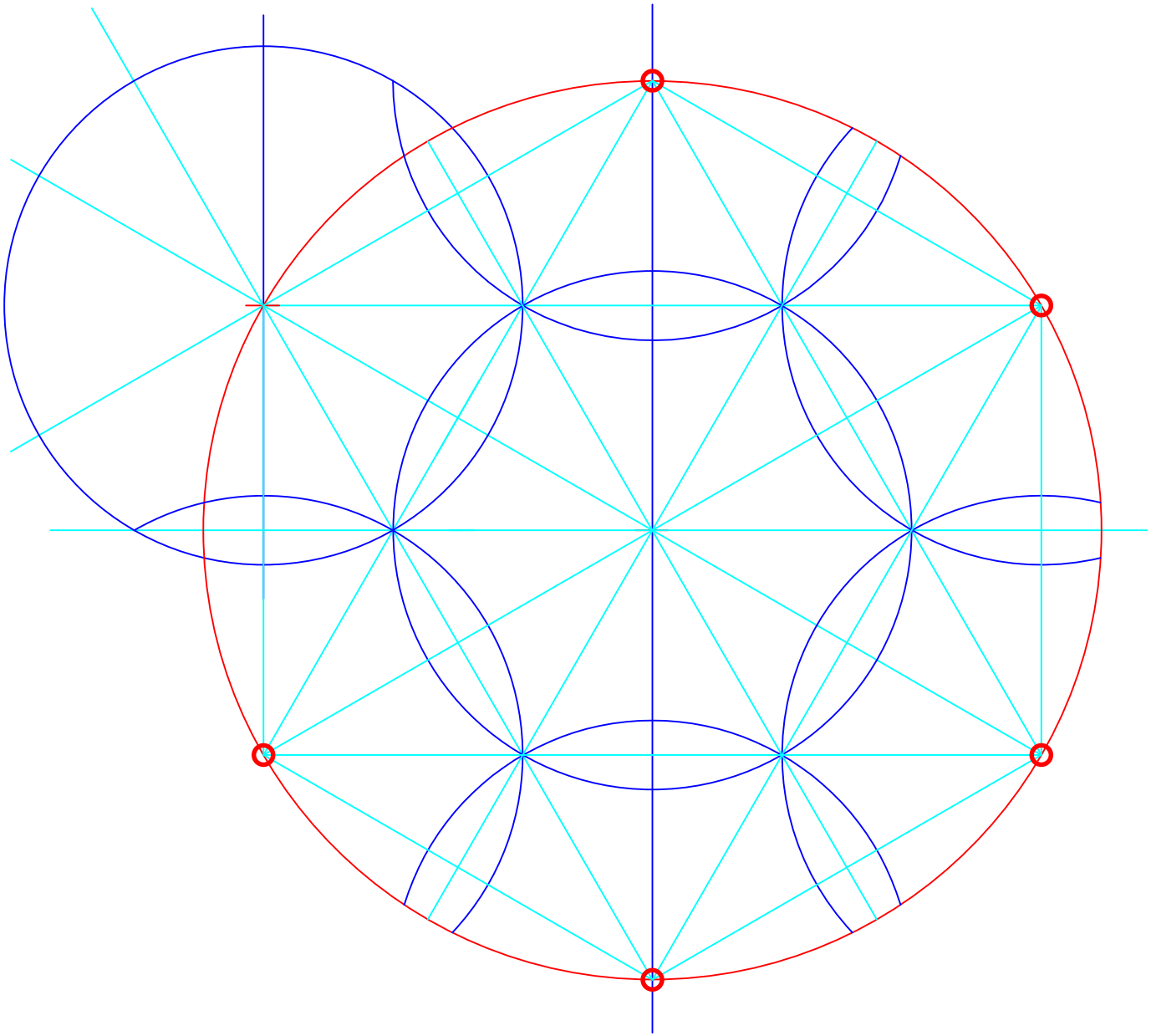
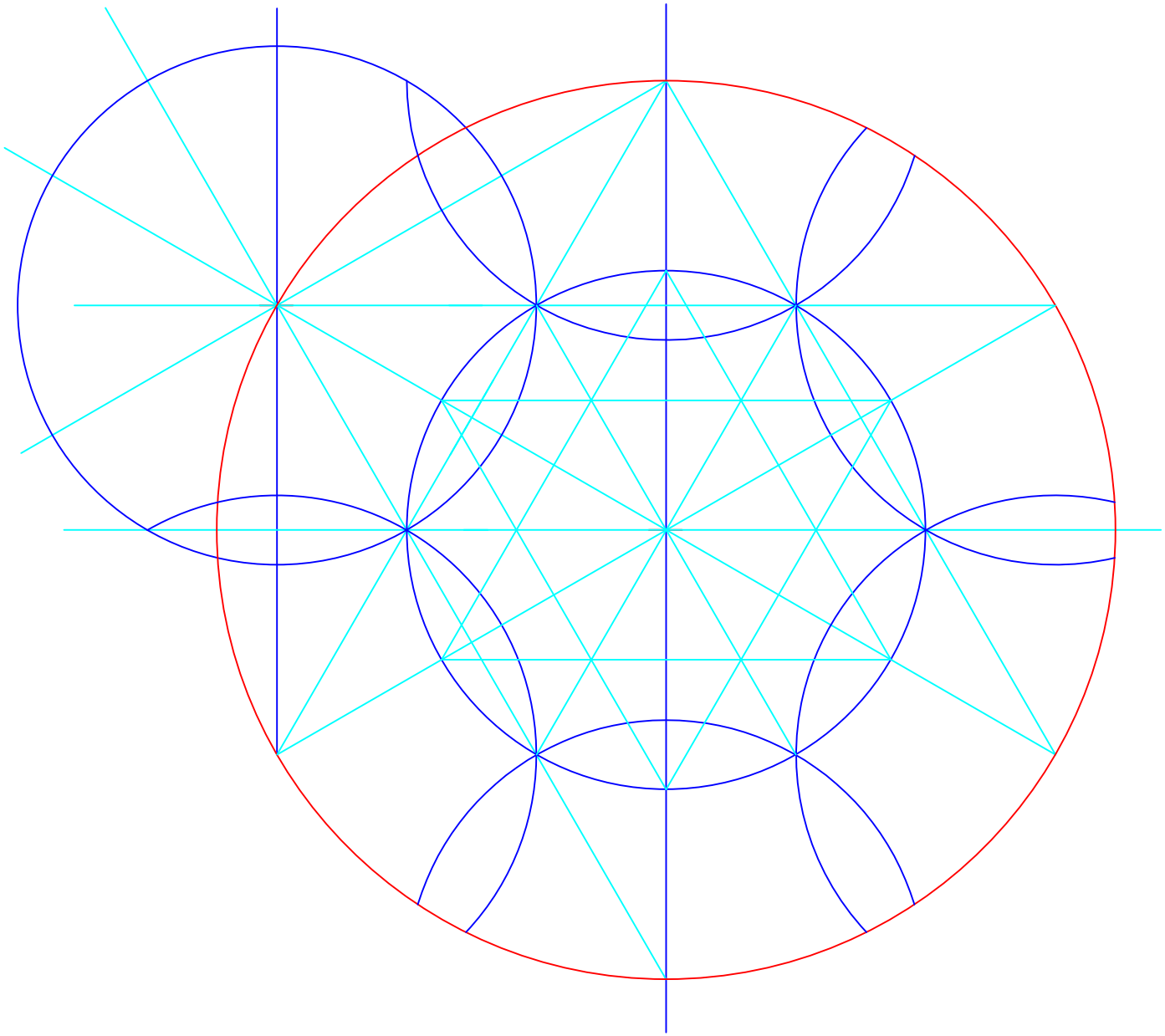
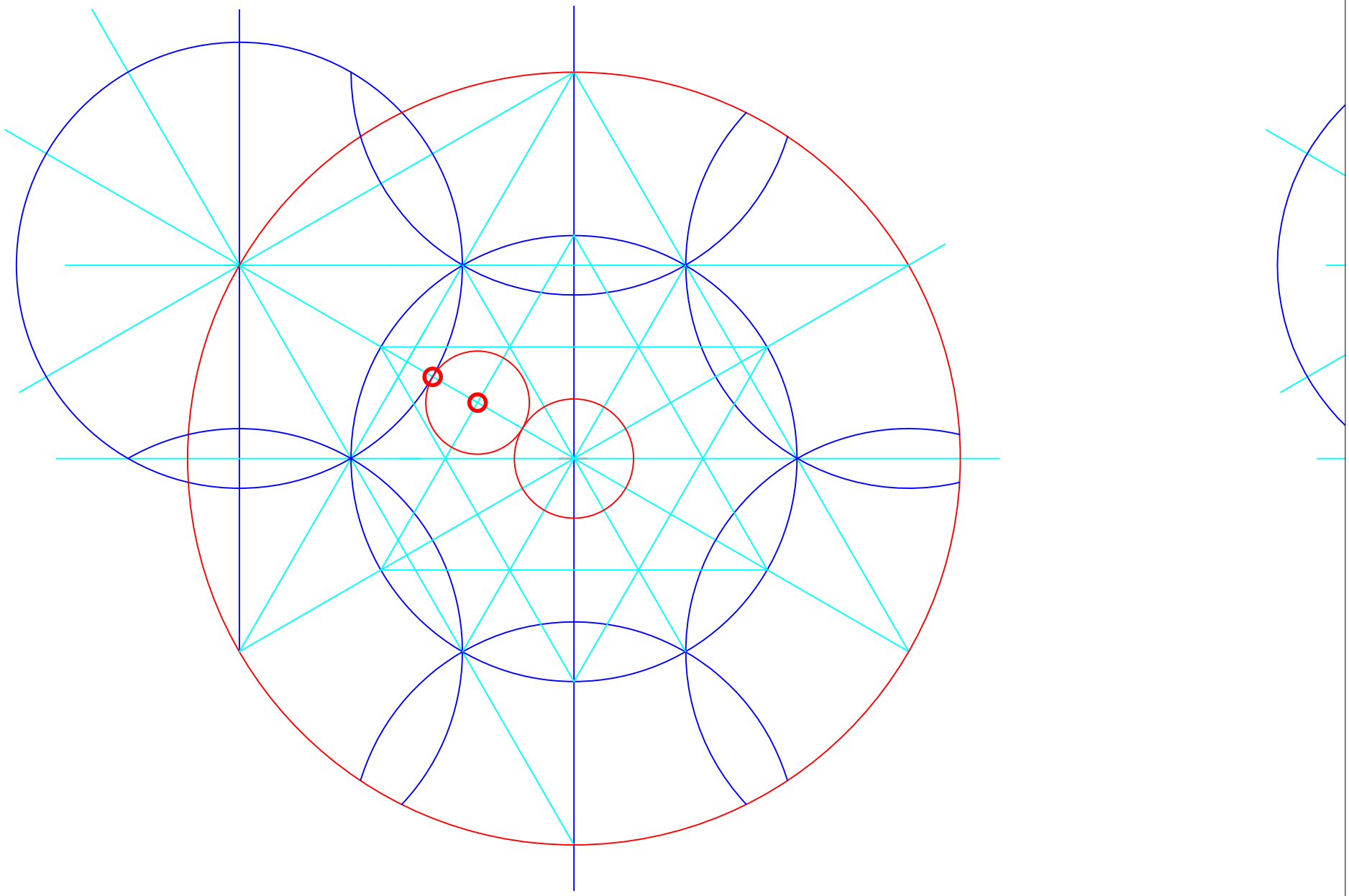
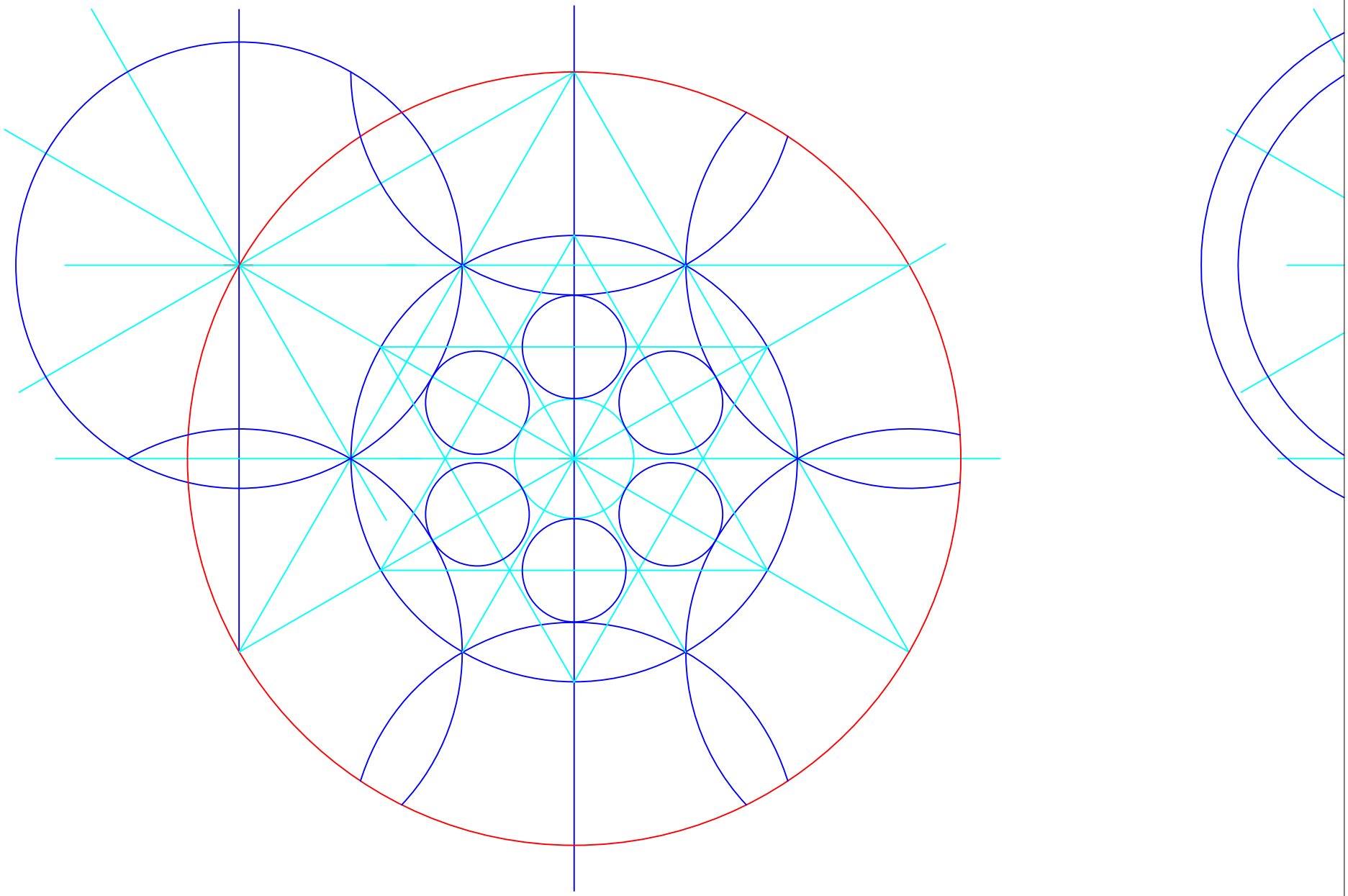
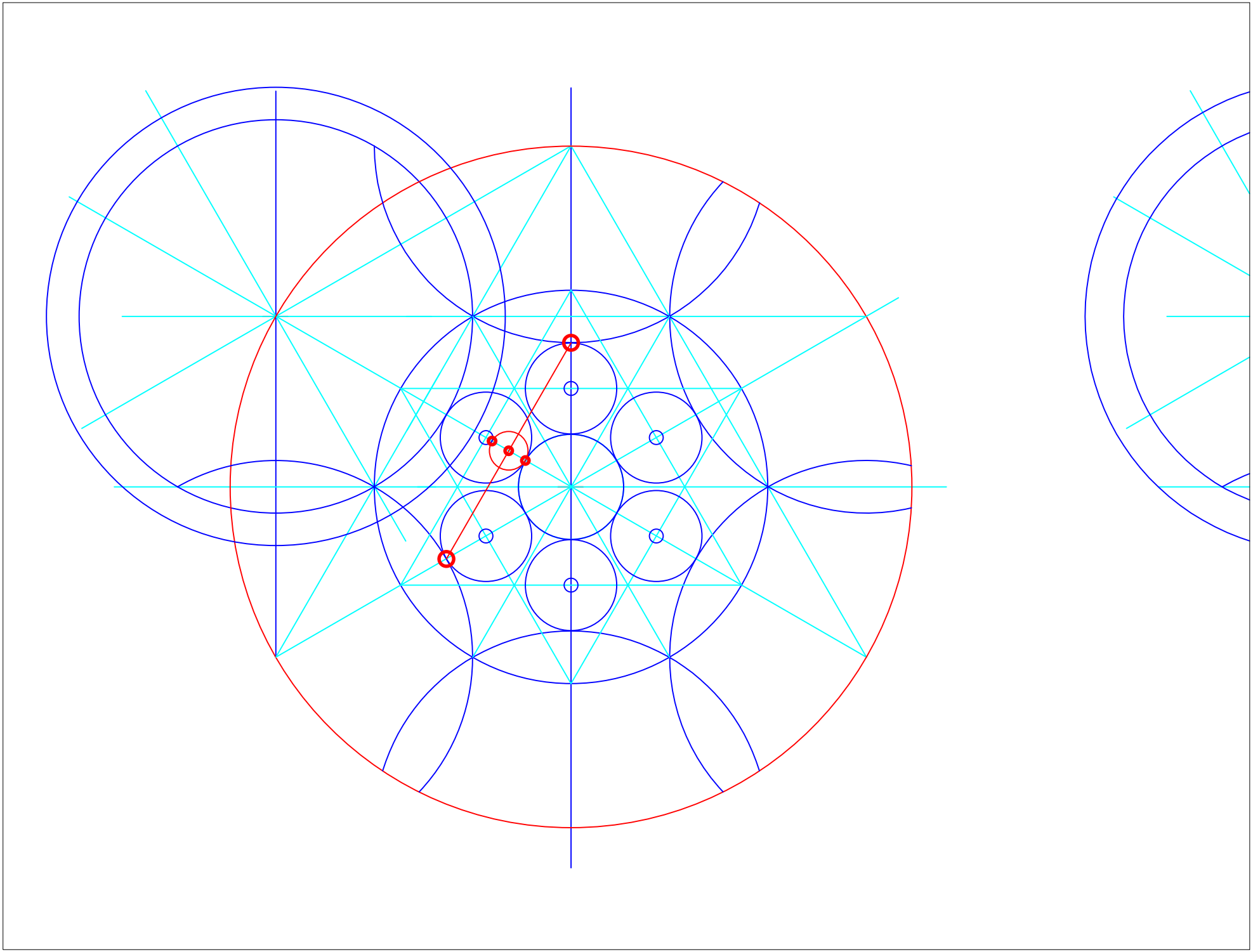
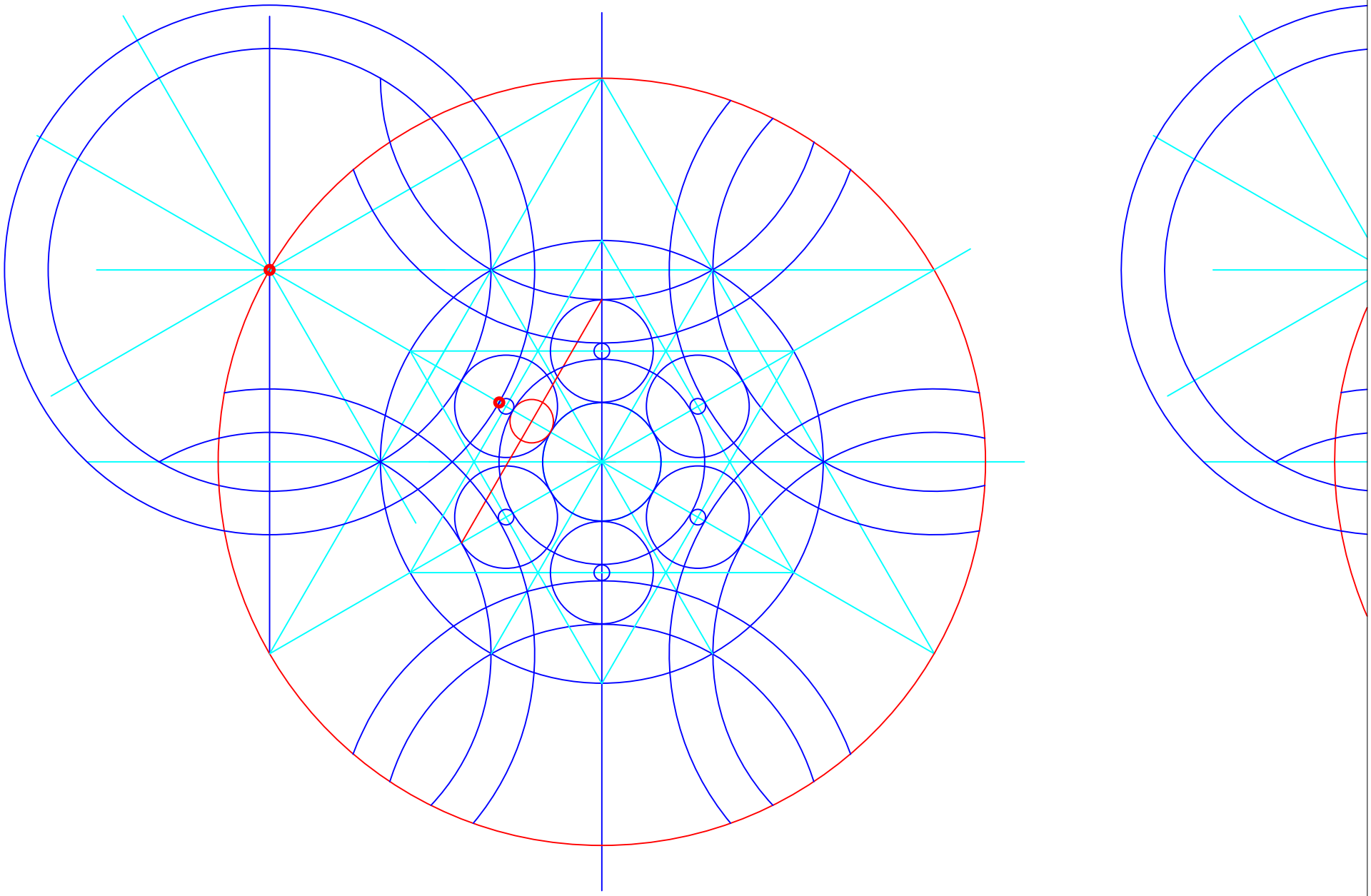
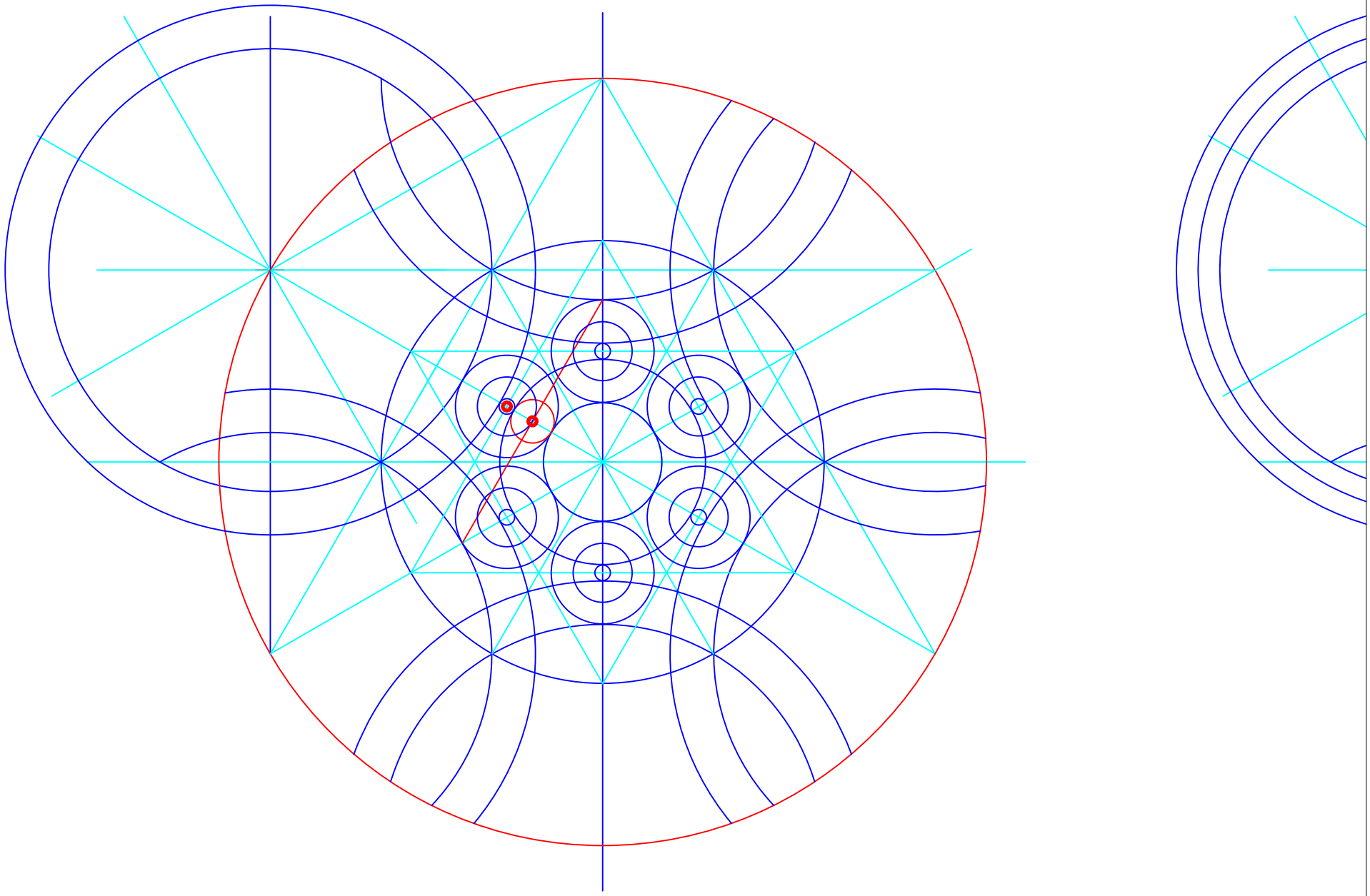
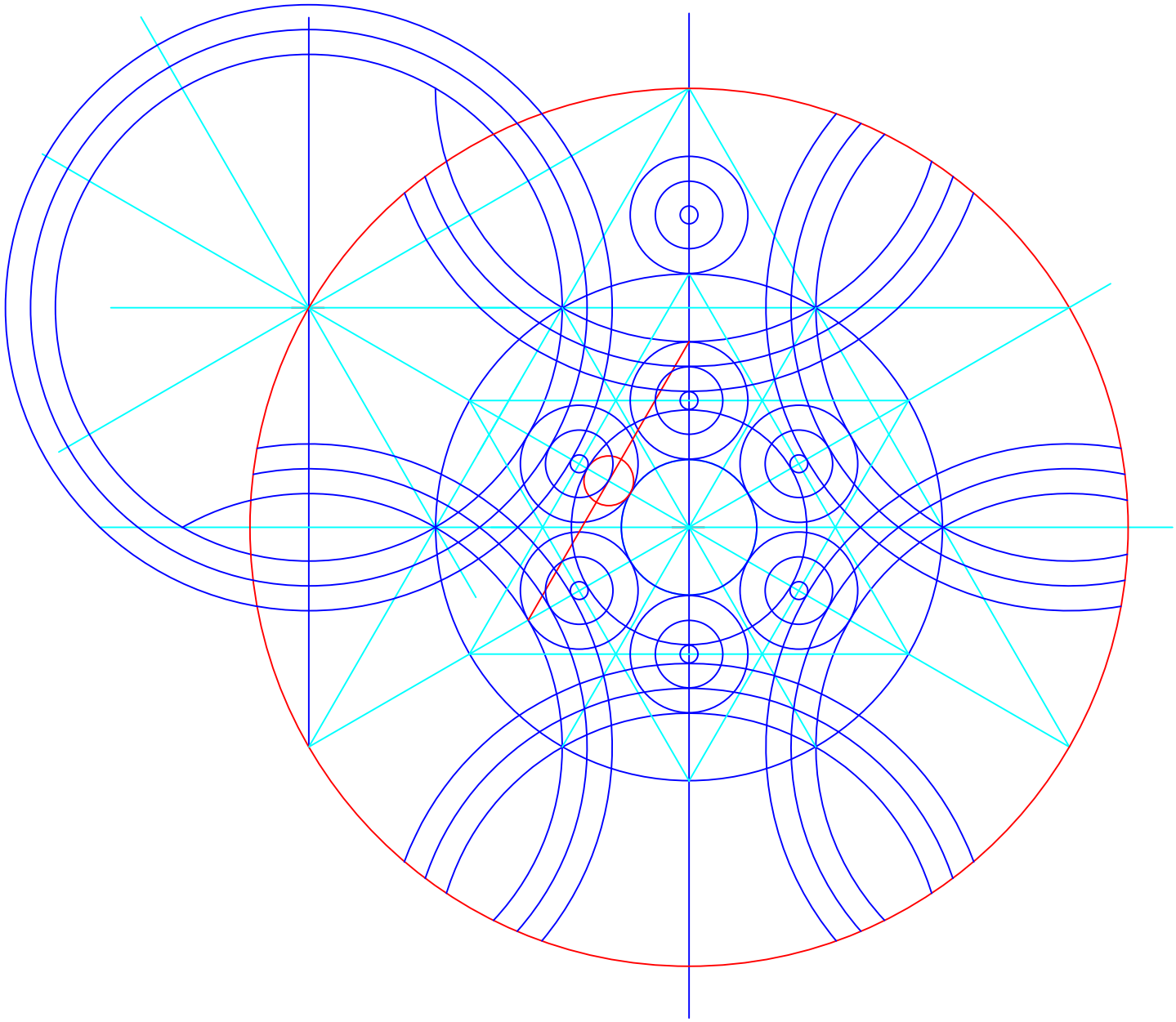
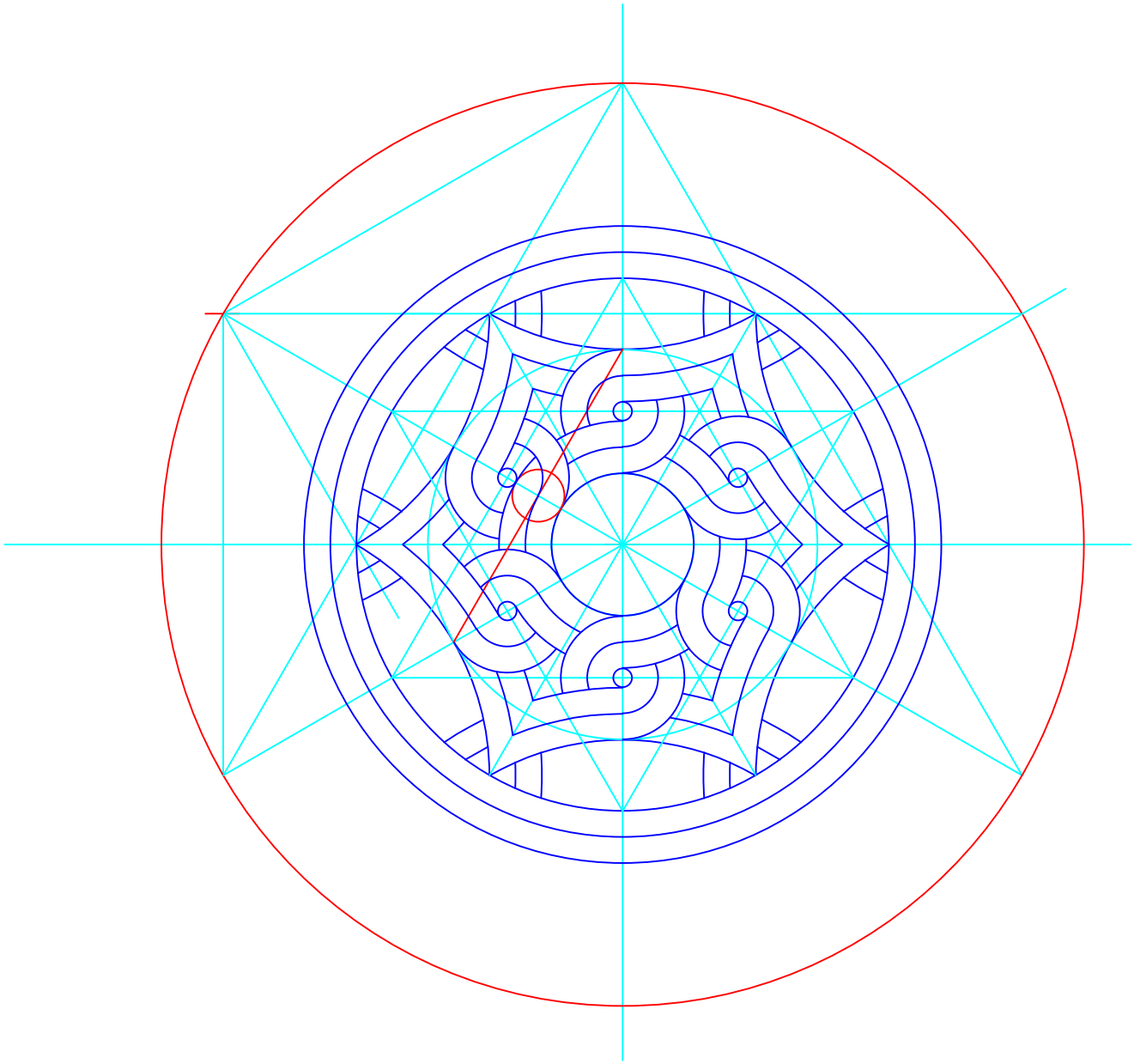
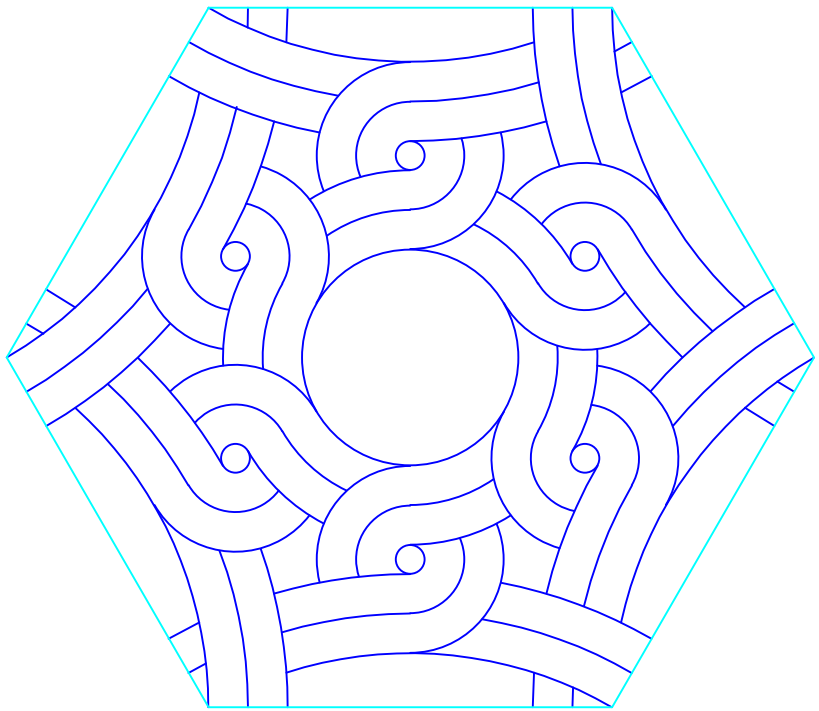
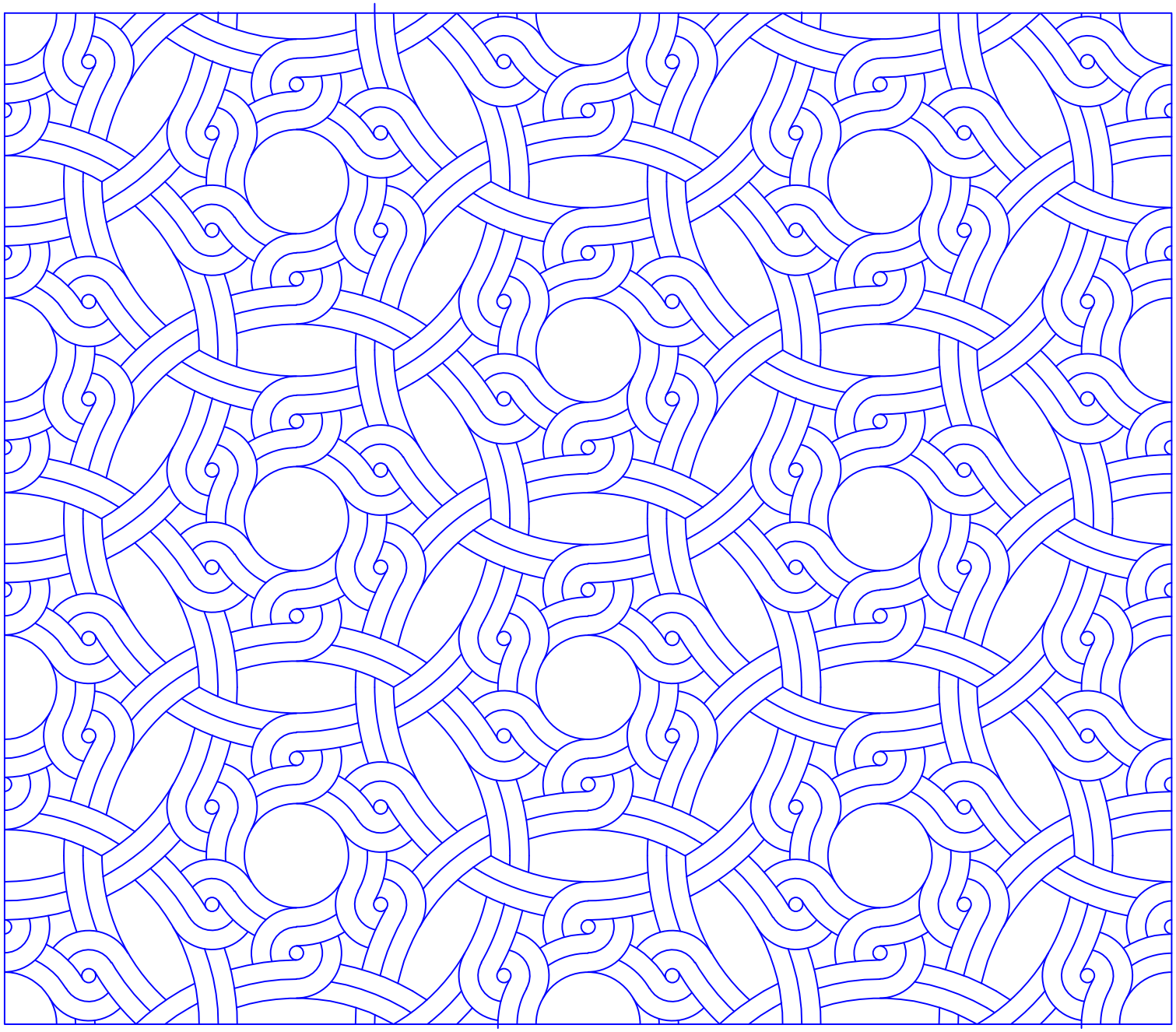
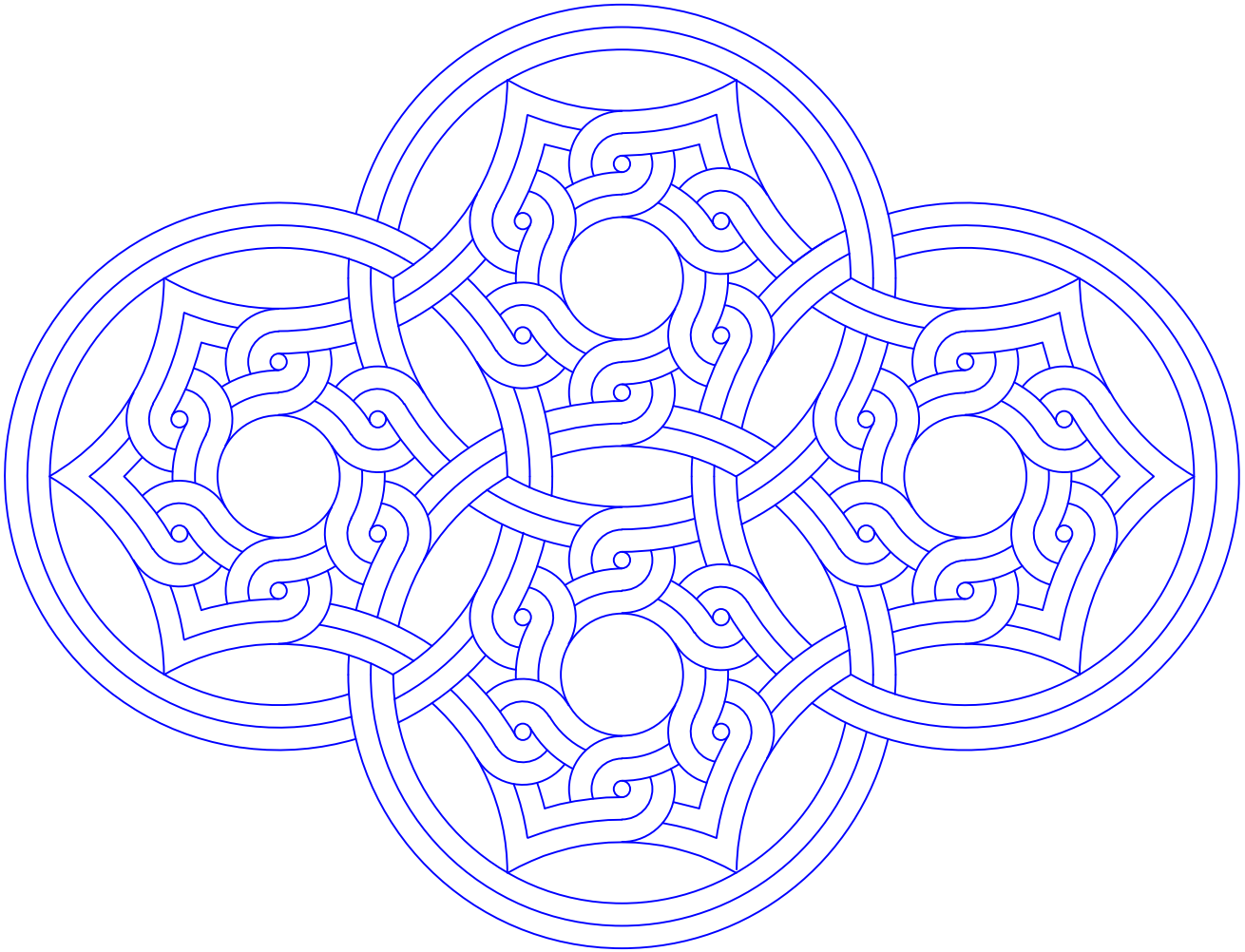


The Khirbat al Mafjar oculus as a repeating pattern. This pattern is identical to the oculus.

.

This serves as a nice introduction to the Early Islamic patterns which are still closely related to the Byzantine tradition- and are still unique.

.

The pattern is relatively simple to construct but it shows most of the problems which need to be solved for tangent circle patterns.

Alan D Adams, Holland New York, Mar 2017. CC BY 4.0, except as noted Drawings and text. https://creativecommons.org/licenses/by/4.0/